Trade Liberalization and Labor-Market Outcomes: Evidence from US Matched Employer-Employee Data

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Abstract

We use matched employer-employee data to examine outcomes among workers initially employed within and outside manufacturing after trade liberalization with China. We find that exposure to this shock operates predominantly through workers' counties (versus industries), that larger own industry and downstream exposure typically reduce relative earnings, and that greater upstream exposure often raises them. The latter is particularly important outside manufacturing: while we find substantial and persistent predicted declines in relative earnings among manufacturing workers, those outside manufacturing are generally predicted to experience relative earnings gains. Investigation of employment reactions indicates they account for a small share of the earnings effect.

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1 Introduction

A large literature in labor economics starting with Podgursky and Swaim (1987) and Jacobson et al. (1993) examines the immediate and long-term consequences of involuntary displacement on worker earnings and employment. A related literature in international trade investigates the impact of import competition on labor demand across a wide range of economies, including India (Topalova, 2007), Brazil (Dix-Carneiro and Kovak, 2017), Canada (Kovak and Morrow, 2022) and Finland (Costinot et al., 2022). Of particular recent interest is the sharp response of US workers (Autor et al., 2014), firms (Greenland et al., 2020), industries (Pierce and Schott, 2016) and regions (Autor et al., 2013; Bloom et al., 2019) to trade liberalization with China. Though these papers agree that the effects of the "China shock" were substantial, they differ in their assessment of its influence on worker earnings versus employment, the effects outside manufacturing versus within, and the extent to which workers benefit from greater exposure among their upstream inputs. The latter is particularly important for understanding welfare gains as well as losses (Feenstra et al., 2017).

In this paper, we contribute to the burgeoning "China Shock" literature – which generally focuses on regions as a unit of analysis – by examining outcomes at the worker level using detailed employer-employee data available from the US Census Bureau's Longitudinal Employer-Household Dynamics (LEHD) program. The LEHD is attractive for several reasons. First, it tracks the earnings and employment of nearly the universe of workers among US states participating in the program, allowing us to identify workers with high versus low labor force attachment in the years leading up to the shock. Second, workers can be matched to a rich set of personal and professional characteristics via links to other Census datasets, e.g., worker traits in the Decennial Census (DC) and plant and firm attributes in the Longitudinal Business Database (LBD) and Longitudinal Foreign Trade Transactions Database (LFTTD). Controlling for these characteristics allows us to identify earnings impacts more clearly. Third, the data span virtually all industries, enabling comparison of workers' responses within versus outside manufacturing.

The first part of the paper exploits the comprehensiveness of the LEHD to describe unconditional US worker flows and earnings trajectories from 2000 to 2007, a period of steep manufacturing employment decline from the US extension of Permanent Normal Trade Relations (PNTR) to China to the onset of the Great Recession. This exercise yields a number of facts that provide useful context for the "China Shock". First, we show that despite the substantial -3.5 million net decline in manufacturing employment between 2000 and 2007, gross flows into manufacturing are large, with 3.8 million workers transitioning into this sector from services, and 2.4 million from non-employment. Second, we find that remaining in manufacturing is by far the most prevalent outcome of those initially employed in that sector, and that median nominal earnings growth among these workers is 27 percent over the period, or about 3 percent per year. Third, the most popular non-manufacturing destination both in levels and relative to initial size is Wholesale (NAICS 42). Workers following this path exhibit median nominal earnings growth slightly higher than those staying in manufacturing, at 33 percent over the period. This trend may reflect workers remaining in the same occupation as they switch industries (Traiberman, 2016), e.g., a logistics specialist moving from a manufacturing plant to a disribution center, as well as former manufacturing plants or firms segueing into factoryless goods production

(Fort, 2017; Ding et al., 2019; Bloom et al., 2019).

Two alternate destinations of note are Construction (NAICS 23) and Administrative and Support Services (NAICS 56), which includes staffing agencies that might inherit workers from outsourcing manufacturing firms (Dey et al., 2012). These sectors are in the middle of the pack in terms of flow adjusted for initial size, accounting for 0.5 and 4.2 million gross transitions, respectively. Construction has been suggested as a relatively high-paying haven for those with manufacturing skills (Charles et al., 2016) and, though the flow is relatively small, we do indeed find that manufacturing workers moving into construction experience median earnings growth of 30 percent. Workers making the transition to Administrative and Support Services, by contrast, have the lowest observed median nominal earnings growth, at -22 percent over the period. Earnings growth is also negative for those moving from manufacturing to Accommodation and Food (NAICS 72), and just slightly positive for those heading to Retail (NAICS 44-5), both of which are consistent with anecdotal evidence that displaced manufacturing workers face large wage declines as they move into lower-skill service sectors (Scott et al., 2022).

In the second part of the paper we use a series of worker-level difference-in-differences (DID) regressions to examine how workers' earnings and employment evolve after versus before PNTR (first difference) based on their exposure to the change in policy (second difference). We focus on workers with continuous employment in the pre-period, breaking them into four groups according to whether they are initially employed within or outside manufacturing, and whether they remain with the same firm for the entire pre-period. We refer to the latter distinction as having high versus low labor force attachment. Given the richness of our data, we are able to consider two forms of exposure to the shock, via industry and via county of initial employment, where the latter is a Bartik-style employment-weighted-average across initial industries produced in each county. These measures allow us to differentiate sectoral (industry) versus spatial (county) frictions to reallocation, and their potential interaction. All specifications include worker and year fixed effects, as well as interactions of a broad set of initial worker and firm characteristics with a post-PNTR dummy. The latter allow the relationships between these attributes and earnings to change at the same time that PNTR was granted, which assists us in isolating effects due to the policy change.

In our baseline specification we follow most of the literature and restrict attention to estimating the impact of own-industry and own-county exposure, which emphasizes the negative consequences of import competition. Three findings stand out. First, we show that for manufacturing workers, exposure to PNTR operates predominantly through their county rather than industry, highlighting the predominance of binding geographic reallocation costs.[1] Second, we find that estimated county exposure coefficients are negative, and roughly similar in magnitude and statistical significance, across all four groups of workers, with interquartile increases in own-county exposure implying relative drops in post-period earnings of -28 and -25 percent among high- and low-attachment workers initially employed in manufacturing, and -28 to -40 percent among high- and low-attachment workers initially outside that sector. These results suggest manufacturing and non-manufacturing workers compete in overlapping labor markets within counties. Third, we find that while the pattern of results is similar

[1] Own industry exposure is not defined for non-manufacturing workers.

for employment, the magnitude of the estimated effect is much smaller, suggesting most of the relative declines in earnings are due to relative drops in wages.

In our second, preferred specification, we consider workers' exposure to PNTR along the value chain. To do so, we include industry and county upstream and downstream exposures in addition to the baseline own-industry and own-county exposures. Up- and downstream exposure for industry j are defined as input-output-coefficient weighted averages of the exposure of j's supplier and customer industries. Likewise, county up- and downstream exposures are constructed as the average up- and downstream exposure of the industries present in a county, weighted by those industries' employment.

Results from our preferred specification reveal that our baseline findings mask substantial heterogeneity in manufacturing versus non-manufacturing workers' sensitivity to the trade liberalization once exposure via input-output linkages is considered. As before, we see that the impact of the policy change operates overwhelmingly through geographic exposure, as only the coefficient estimates for the county exposures tend to be statistically significant. For all four groups of workers, for both earnings and employment, however, we find that coefficients of county upstream exposure are positive, while those for county own and downstream exposure are negative. These results demonstrate that greater county upstream exposure can partially or completely offset higher county own or downstream exposure, though this ability varies by sector. For workers outside manufacturing, all three county exposures are individually significant. Among manufacturing workers, however, coefficients on county own and upstream exposure are generally small in magnitude and not statistically significant, while those on county downstream exposure are relatively large in magnitude and always statistically significant.

In terms of economic significance, we find that differences between the baseline- and preferred-specification estimates yield sharply divergent predictions for pre- versus post-PNTR earnings growth among manufacturing versus non-manufacturing workers. We demonstrate this disparity at the county level by multiplying county exposure coefficients by counties' actual exposures. For initial non-manufacturing workers, counties' predicted relative earnings growth switches from universally negative under the baseline to generally positive using the preferred specification estimates. Indeed, under the preferred specification, predicted relative earnings growth is positive in 90 percent of counties among high-attachment non-manufacturing workers. For low-attachment non-manufacturing workers, the share is 58 percent. Factoring in upstream exposure also mitigates the relative drop in predicted earnings among manufacturing workers, but rarely results in a positive prediction. For these workers, just 3 and 1 percent of counties are predicted to have relative earnings growth under the preferred specification. These differences in outcomes for non-manufacturing versus manufacturing workers highlight the substantial benefits that accrue to non-manufacturing workers arising from trade liberalization on the inputs used in their industries.

An interesting example highlighting these differences is Napa County, California. Napa has a relatively high county upstream gap but a relatively low county downstream gap: its mix of sectors use inputs that are heavily exposed to PNTR's trade liberalization, but they generally sell to consumers. As a consequence, its baseline versus predicted relative earnings growth among high-attachment non-manufacturing workers jumps from a drop of -20 percent under the baseline to an increase of 25

percent under the preferred, and from drops of -30 percent to -1 percent among those with low-attachment. The analogous changes among high- and low-attachment manufacturing workers are impressive, from -21 to -12, and -18 to -12 percent, but do not reverse sign. San Mateo County, California, also known as Silicon Valley, serves as a useful counter-example, having relatively high downstream exposure and relatively low upstream exposure. In this county, predicted relative earnings growth for manufacturing workers is lower under the preferred specification even though it is higher for non-manufacturing workers.

The disparity between our baseline and preferred specifications are particularly noteworthy as the few existing studies of the "China Shock" that do include exposure via input-output linkages, e.g., Pierce and Schott (2016) and Autor et al. (2016), find no evidence of a pro-competitive benefit from trade liberalization. Our results here suggest examinations of trade liberalizations without consideration of both up- and downstream exposure may be incomplete.[2]

In the final part of the paper, we investigate heterogeneous responses along a rich set of worker and firm attributes, many of which have not been examined for US workers. While we currently examine these responses using our baseline specification, we plan to include estimates using our preferred specification in an upcoming draft. Our most consistent result across the four groups of workers we study relates to gender, were we find that the impact of own-county exposure on pre- versus post-PNTR earnings and employment is less negative for females than males, though the earnings results are statistically significant only for workers initially employed outside manufacturing. This trend may reflect labor-market segmentation, e.g., displaced male manufacturing worker wages competing disproportionately with males outside manufacturing. Such an outcome would be consistent with the disproportionately negative effect of Chinese import competition on male socioeconomic outcomes documented in Autor et al. (2019) and Pierce and Schott (2020).

Examining outcomes by workers' initial education and level of income, we find some evidence that low-attachment, non-manufacturing workers with more education have relatively better earnings and employment outcomes than those with less education, which is consistent with previous research (Autor et al., 2013). However, among initial manufacturing workers, we find the worst earnings and employment outcomes for those with the *highest* initial incomes, whereas Autor et al. (2014) find larger earnings losses for initially low-wage workers. This novel result may reflect the loss of firm- and industry-specific human capital that is not easily transferable outside the manufacturing sector, reminiscent of the plight of managers laid off in the Pennsylvania steel industry in the 1970s and 80s (Sullivan and Wachter, 2009).

Our analysis of the implication of firm attributes for US workers' adjustments to trade shocks is, to our knowledge, the first in the literature, as it is only possible with linked employer-employee data such as the LEHD.[3] We find some advantages for workers at trading firms that respond to import

[2]Using commuting-zone-level data, Autor et al. (2013) find that greater own-region exposure to imports from China has no impact on manufacturing wages but does lower non-manufacturing wages. Exploiting worker-level social security data, Autor et al. (2014) find that greater own-industry and own-commuting-zone exposure to China reduces cumulative earnings among manufacturing workers. Hakobyan and McLaren (2016) find that own-industry and own-county exposure to NAFTA is associated with substantial wage declines among less educated workers in both manufacturing and non-manufacturing.

[3]As discussed below, others, such as Krishna et al. (2014) have considered the effects of firm attributes using data

competition by taking advantage of the potential benefits of trade liberalization. Specifically, we find that workers at firms that only import tend to have relatively better post-PNTR earnings than workers at firms that do not import. Importantly, this response occurs not only due to relatively better performance for earnings, but also for employment. The lower probability of experiencing PNTR-related non-employment for workers at importing firms indicates that those workers are shielded from one of the most disruptive potential effects of trade liberalization.

Our characterization of worker earnings and employment before and after a large shock to labor demand contributes most directly to the literature using US Social Security Administration (SSA) data to investigate the short- and long-run consequences of "mass layoffs," typically defined as separation by workers with three to six years tenure from an establishment shedding 30 percent or more of its labor force within a year. Papers in this line of research generally find that earnings drop 30 to 40 percent upon displacement before staging a modest but often incomplete recovery in the subsequent decade (Jacobson et al., 1993; Von Wachter et al., 2009). Here, rather than infer mass layoff we exploit a plausibly exogenous change in US trade policy as a potential casuse of it.

Building on this work, a long line of research exploits the labor demand shocks associated with international trade to consider effects on a wide range of employment responses, with recent research increasingly exploiting worker-level data.[4] In the analysis most closely related to ours, Autor et al. (2014) find that over the period examined in this paper, workers initially employed in import-competing manufacturing industries exhibit disproportionately large losses in cumulative earnings. Hakobyan and McLaren (2016) use publicly available data from the US Census Bureau's IPUMS dataset to document a decline in wages of 8 percentage points among US manufacturing and non-manufacturing workers in industries and regions with greater exposure to import competition from Mexico following NAFTA, and Kovak and Morrow (2022) estimate the effects on worker earnings of both export expansion and import competition following CUSFTA. Our examination of the relevance of firm characteristics in workers' response to trade policy relates to Krishna et al. (2014) who find that workers at Brazilian exporting firms do not experience differential changes in wages relative to workers at non-exporting firms. In addition, Deng et al. (2021) leverage German employer-employee data to explore differences in effects of industry- versus occupation-level exposure to import competition on income risk. Our contribution relative to these efforts is to consider impacts for US workers, to measure exposure to the trade shock via geographic labor market as well as industry, and to evaluate the influence of exposure along the value chain as well as a wide range of worker and firm attributes on worker outcomes.

The remainder of the paper proceeds as follows. Section 2 summarizes the matched employer-employee data we use. Section 3 provides a detailed accounting of gross manufacturing employment in- and outflows between 2000 and 2007. Section 4 describes the trade liberalization we study. Sections 5 and 6 present our main empirical results. Section 7 concludes.

for other countries.

[4]This literature is succinctly surveyed in McLaren (2017), McLaren (2022), and Caliendo and Parro (2022).

2 US Employer-Employee Data

We examine the relationship between US worker outcomes and exposure to PNTR using longitudinally linked employer-employee data from the US Census Bureau's Longitudinal Employer-Household Dynamics (LEHD) program, created as part of the Local Employment Dynamics federal-state partnership. The earnings and employment data are derived from state unemployment insurance (UI) records and the Quarterly Census of Employment and Wages (QCEW). In each quarter in each state, firms subject to state UI laws submit the earnings of their employees to their UI program, where earnings are defined as the sum of gross wages, salaries, bonuses and tips.[5]

States match the firm identifiers in these records to the QCEW, which contains information about where the firms are located and their industries of activity, and pass these data to the US Census Bureau. Census adds information about workers' age, gender, race, birth country and educational attainment derived from several sources, including the Decennial Census. This information is collected in the LEHD's Individual Characteristics File (ICF).[6] Birth country is either US or foreign. Racial categories are White, Black, Asian and Other. Education attainment levels are less than high school, high school or the equivalent, some college, and bachelors degree or higher.[7]

Census uses several levels of firm and establishment identifiers across various datasets. Firms in the LEHD are identified by state employer identification numbers (SEINs). Concordances between SEINs and Census' other identifiers allow us to match workers in the LEHD to firm-level data contained in the Longitudinal Business Database (LBD), which tracks employment and other attributes of virtually all privately owned firms in the United States. Via the LBD, we are able to measure the size of a worker's firm as well as whether the firm has multiple establishments.

In any given year a worker may be employed by more than one firm. We adopt the convention among LEHD users of assigning each worker in each year to the firm at which the workers' earnings are highest. Firms can have multiple establishments, and these establishments can have different six-digit NAICS industry codes and be located in different counties within the state.[8] We assign workers to establishments within the firm (and, thereby industries and counties) using the firm-establishment imputation in the LEHD's Unit-to-Worker (U2W) file.

As illustrated in appendix Figure A.1, the number of states for which data are available in the LEHD varies over time. In the difference-in-differences estimations we present in Section 5, we use stratified random samples drawn from the 19 states whose information is available for our full pre- and post-PNTR sample period, 1993 to 2014.[9] For the descriptive results on workers' industry switching

[5] As discussed in greater detail in Abowd et al. (2009) and Vilhuber and McKinney (2014), state UI records cover approximately 96 percent of all private sector employees as well as the employees of state and local governments. Prime exceptions are agriculture, self-employed individuals and some parts of the public sector, in particular federal, military and postal workers.

[6] Workers in the LEHD are identified via anonymous longitudinal person identifiers (PIKs) which have a one-to-one correspondence with their social security numbers and which are used to identify workers in a range of Census datasets. Except for Minnesota, UI records do not contain any information about firms except their identifier.

[7] Note that educational attainment is imputed for the majority of the sample. Overall for the base LEHD sample, about 92% of PIKs have education imputed. See Vilhuber and McKinney (2014) for more details.

[8] We use the updated "FK" NAICS industry identifiers provided by Fort and Klimek (2016).

[9] The 19 states are Alaska, Arizona, California, Colorado, Florida, Idaho, Illinois, Indiana, Kansas, Louisiana, Maryland, Missouri, Montana, North Carolina, Oregon, Pennsylvania, Washington, Wisconsin, and Wyoming. They represent

over time, in Section 3, however, we use information from the 46 states whose data are in the LEHD starting in 2000.[10]

For our regression analysis, we create two random samples designed to capture two different levels of workforce attachment and potential possession of firm-specific human capital. The "high-attachment" sample starts with all workers from the 19-state sample that are employed continuously by the same firm from 1993 to 1999. The "low-attachment" sample, by contrast, begins with all workers employed in each year from 1993 to 1999, irrespective of whether they are employed by the same firm throughout. For computational tractability, from each of these subsets, we draw a random 5 percent sample stratified to be representative of the full LEHD. This draw includes all workers from "small" counties (i.e., those in the first size decile, with population at or below 5327 according to the 2000 census), plus a 5 percent random sample of workers from all other, i.e., "large", counties, stratified according to worker attributes (age, gender, race, ethnicity and educational attainment). Note that all of our regressions will be weighted by the inverse of the probability of being in the sample. Finally, we eliminate workers from this draw who will be older than the retirement age (64) in 2014.

Within each sample, workers are classified as initially in manufacturing (M) if they are employed in an establishment whose major activity in 1999 is in NAICS industries beginning with "3". All other workers are classified as non-manufacturing (NM). Workers who appear in only some years of our sample are classified as not employed (NE) in the remaining years. The predominant reason for NE status is lack of employment—unemployment or labor force exit—but it may also be the result of death, movement to a state (or country) outside the sample of states for which we have data, or movement to a job that is out of scope of the UI system.

3 Post 2000 US Labor Reallocation

In this section, we provide context existing research on the employment effects of the China Shock, and, more generally, on US structural change (Ding et al., 2019), by summarizing workers' post-PNTR transitions among sectors, and the earnings growth associated with these moves. While straightforward, these descriptions – to our knowledge – provide the first detailed accounting of sector-to-sector flows for manufacturing workers during this period, and therefore offer a more complete view of the labor market transitions of manufacturing workers at the onset of the steep increase in import competition from China.

3.1 Transitions Among M, NM and NE

First, we provide a broad overview of workers' gross flows among manufacturing (M), non-manufacturing (NM) and non-employment (NE) from 2000 to 2007 in Table 1 using the 46-state sample described

47 percent of US overall and manufacturing employment in 2000.

[10] Missing from the 46-state sample are Alabama, Arkansas, New Hampshire, Mississippi and the District of Columbia. These states represent 96 and 95 percent of US overall and manufacturing employment in 2000.

in Section 2.[11] To isolate the impact of retirement, we restrict the sample to workers below the age of 50 in the year 2000. The top panel reports these flows in millions of workers, while the bottom panel expresses them as percentages of origin sectors' initial levels. As indicated in the upper panel, the number of manufacturing workers declines from 17.5 million in 2000 (row 2, final column) to 14.0 million in 2007 (column 2, final row), while non-manufacturing employment decreases from 117.6 to 116.0 million.[12]

Table 1 reveals two novel and interesting features of labor-market adjustment in the post-PNTR period. First, we see that even as many workers left manufacturing, employment declines were partially offset by sizable inflows from industries outside the sector. From 2000 to 2007, 3.8 million workers move from NM to M, and 2.4 million transition from NE to M, with the result that 44 percent of workers working at a manufacturing establishment in 2007 were not employed at such a plant in 2000.[13] Thus, there is substantial switching *into* manufacturing, even in a time of precipitous net decline.

The second noteworthy trend in Table 1 is that, despite the steep decline in manufacturing employment and associated negative socioeconomic implications discussed in the literature (Feler and Senses, 2017; Autor et al., 2019; Pierce and Schott, 2020), the share of under-50 year 2000 employees transitioning to non-employment in 2007 is similar for manufacturing and non-manufacturing. As shown in the lower panel of Table 1, 23 percent of 2000 M workers transitioning to NE in 2007, versus 25 percent for NM workers. One potential explanation for this similarity is that PNTR contributed to displacement of workers initially employed within or outside manufacturing. Support for this mechanism is suggested by our regression results in Section 5, which show that worker exposure to PNTR is transmitted primarily through workers' local labor markets (i.e., counties) rather than their industry of employment.[14]

[11] The analysis ends in 2007 to focus on worker reallocation prior to the Great Recession. In Appendix Table A.2 we find that while the general pattern of movement is similar for the periods ending in 2005 and 2011, there is greater transition away from initial sector over longer intervals.

[12] Over the same interval, i.e., from December 1999 to December 2007, US labor force participation drops from 67.1 to 66.0 percent according to the US Bureau of Labor Statistics. According to the www.bls.gov, US manufacturing employment falls from 16.8 to 13.7 million, while overall employment rises from 112 to 139 million.

[13] One source of flows from NE to M could be the first-time entry of young workers to the labor force.

[14] While these summary statistics are informative, they lack a comparison to transitions in a pre-PNTR period. If manufacturing workers in the pre-PNTR period had been less likely to move to NE (i.e. had higher labor force attachment) than non-manufacturing workers, the roughly even transition rates in the post-PNTR period could indicate a break in trend. While we cannot extend the 46-state sample back in time, we plan to include comparable statistics for a smaller set of states over a suitable pre-period in a future draft.

Table 1: Major Sector Transition Matrix, 2000-7

	Employment (Millions)			
	Sector in 2007			Total in
Sector in 2000	NM	M	NE	2000
Non-Manufacturing (NM)	84.8	3.8	29.0	117.6
Manufacturing (M)	5.7	7.8	4.0	17.5
Not Employed (NE)	25.5	2.4	.	27.9
Total in 2007	116.0	14.0	33.0	162.9

	Percent of Initial Level			
	Sector in 2007			Total in
Sector in 2000	NM	M	NE	2000
Non-Manufacturing (NM)	72	3	25	100
Manufacturing (M)	33	45	23	100
Not Employed (NE)	92	8	.	100
Total in 2007	71	9	20	100

Source: LEHD, LBD and authors' calculations. Table reports the transition paths of employed and not-employed workers from 2000 (row) to 2007 (column) for the 46 states whose information is available in the LEHD starting in 2000. Counts are restricted to workers below the age of 50 in 2000.Alabama, Arkansas, New Hampshire and Mississippi as well as the District of Columbia are excluded. Upper panel reports levels in millions of workers. Lower panel reports shares of initial levels. Appendix Table 1 reports analogous statistics for 2000 to 2005 and 2000 to 2011.

3.2 Detailed Decomposition of Gross M Outflows

We provide a more detailed description of worker reallocation *away* from manufacturing in Table 2, which decomposes manufacturing workers' *gross outflows* from 2000 to 2007 by two-digit NAICS category. The first two columns, which report the level and share of outflows by destination sector. reveal that 44 percent of workers remain in manufacturing. The three largest outflows are towards Wholesale (NAICS 42), Retail (NAICS 44-5) and Administration, Support, and Waste Management (NAICS 56) – hereafter ASWM – accounting for 4.8, 4.3 and 4.2 percent of the total gross outflow.

Table 2: Detailed 2000-7 $M \rightarrow NM$ Transitions (46-States)

	2000-7 Gross Flow Out of Manufacturing		2000 Employment		
	(1)	(2)	(3)	(4)	(5)
Destination NAICS Sector	Flow	% of Flow	Level	% of Total	(2) / (4)
11 Agriculture,Fish,Forest	78	0.5	1,719	1.4	0.33
21 Mining	57	0.3	536	0.4	0.76
22 Utilities	47	0.3	569	0.5	0.59
23 Construction	500	2.9	7,966	6.4	0.45
31-33 Manufacturing	7,780	44.9	17,551	14.1	3.19
42 Wholesale	830	4.8	6,182	5.0	0.97
44-45 Retail	740	4.3	18,172	14.6	0.29
48-49 Transportation	300	1.7	4,602	3.7	0.47
51 Information	120	0.7	4,146	3.3	0.21
52 Finance, Insurance	170	1.0	5,714	4.6	0.21
53 Real Estate, Leasing	100	0.6	2,239	1.8	0.32
54 Professional	470	2.7	7,458	6.0	0.45
55 Management	92	0.5	2,227	1.8	0.30
56 Admin, Support,Waste Mgmt	720	4.2	11,332	9.1	0.46
61 Education	260	1.5	2,031	1.6	0.92
62 Health	480	2.8	13,228	10.6	0.26
71 Arts, Entertain, Recreation	74	0.4	2,236	1.8	0.24
72 Accomodation, Food	310	1.8	12,092	9.7	0.18
81 Other	200	1.2	4,600	3.7	0.31
Not Employed	3,990	23.0			
Total	17,318	13.9	124,598	100.0	

Source: LEHD, LBD, QWI and authors' calculations. First column displays the distribution of initial US employment across the noted sectors in 2000 from the Quarterly Workforce Indicators Database available at https://ledextract.ces.census.gov/static/data.html. Second column reports outflows of manufacturing workers to those sectors between 2000 to 2007 across the 46 states for which information is available in the LEHD in those years (Alabama, Arkansas, New Hampshire Mississippi and DC are excluded). Second two columns of each panel report the flows as a share of initial employment and as a share of the overall outflow from manufacturing.

In column 5, we divide the outflow shares in column 2 by destination sectors' initial employment as a share of the total, in column 4, to assess the *relative* likelihood of former manufacturing workers entering a particular sector. Values of this ratio above unity indicate flows into a sector that are greater than their initial size. Staying in manufacturing remains the most prevalent outcome, and the only destination for which the ratio exceeds 1, at 3.19, reflecting the importance of sector-specific human capital (Neal, 1995; Artuc et al., 2010; Ebenstein et al., 2014; Caliendo et al., 2019). Adjusted for initial size, Wholesale (0.97) continues to be the most popular non-manufacturing destination, while Education (0.92) and Mining (0.76) supplant Retail and ASWM in the remainder of the top three.[15]

Of particular interest in Table 2 is the flow from M to Construction (NAICS 23). Charles et al.

[15] A similar message emerges from Appendix Figure A.4, which provides a different normalization of gross manufacturing outflows by comparing them to gross manufacturing inflows for a slightly different period, 2000 to 2005, and at a slightly higher level of aggregation, one-digit NAICS. As indicated in that figure, the largest net outflows of M employment between 2000 and 2005 are to Not Employed (-.70 million), Business Services (-.60 million), Wholesale, Retail, Transportation and Warehousing (-.50 million), Education and Health (-.42 million), and Mining, Utilities, and Construction (-.22 million).

(2016) suggest workers displaced from manufacturing in the early 2000s may have found a commensurately compensated haven in this sector during the post-2000 housing boom.[16] While the flow of 500 thousand manufacturing workers to Construction ranks fifth in terms of destination sectors in column 2, it is ranked ninth when adjusted for its initial size, just behind Professional, Scientific, and Technical Services (NAICS 54). One reason for Construction's middling performance in the latter likely relates to our finding below that workers' exposure to PNTR comes mainly from their initial county of employment. Feler and Senses (2017), for example, find that higher regional exposure to import competition from China is associated with lower housing prices and demand, dampening the ability of Construction to absorb former M workers unless they move geographically.

Also noteworthy in Table 2 is the flow of 720 thousand workers from M to ASWM (NAICS 56), the largest component of which is staffing services, e.g., temp agencies. Dey et al. (2012) estimate that manufacturing firms' use of staffing-service workers dropped from 2.4 million in 2000 to 1.9 million in 2001, before rising back to 2.3 million in 2006. The initial decline after 2000 may be due to a wave of manufacturing plant closures beginning in 1997 (Fort et al., 2018). The recovery between 2001 and 2007, and the flow depicted in our table, may reflect a new wave of new outsourcing in reaction to PNTR, with a potentially important impact on worker earnings, a subject to which we now turn.

3.3 Earnings Growth Conditional on M Outflow

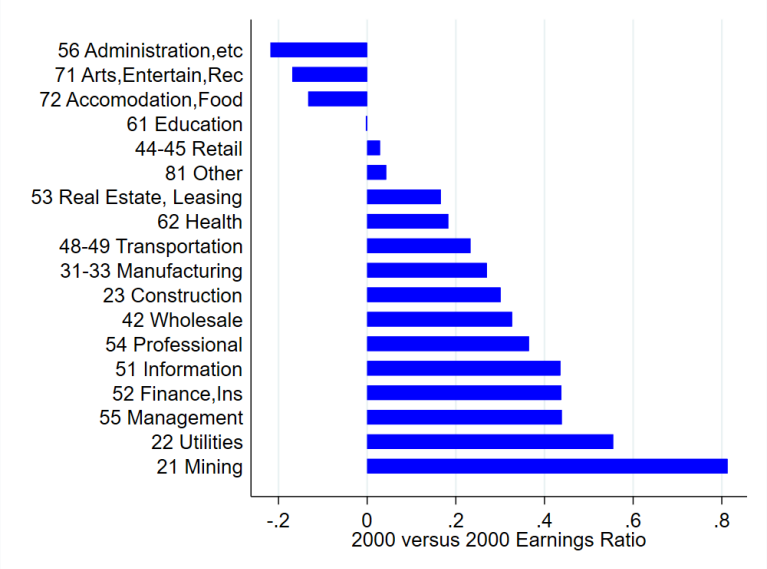
We examine how the earnings of workers initially employed in manufacturing evolve depending upon whether they remain in that sector or migrate to another by calculating the ratio of their 2007 to 2000 earnings. We then take the quasi-median across workers moving to the same destination, and report these medians in Figure 1.[17]

As indicated in the figure, initial M workers experience dramatically different nominal earnings growth depending on their destination sector. For workers who remain in manufacturing, median growth is 27 percent, right in the middle of the pack. Growth is most positive among workers moving to Mining (NAICS 21), Utilities (NAICS 22), and Professional Services (NAICS 51-55), sectors that are intensive in their use of either physical or human capital. Nominal earnings growth is lowest, and actually *negative*, for those transitioning to ASWM (NAICS 56), consistent with the potential increase in outsourcing discussed in the last section. It is also negative for those moving into Arts, Entertainment and Recreation (NAICS 71) and Accommodation and Food Services (NAICS 72), and essentially flat for those heading to Retail (NAICS 44-5). The latter outcomes are consistent

[16] They present three levels of evidence. First, the surge in Construction employment between 2000 and 2006 (933 thousand) appears to partially offset the (3.3 million) decline in manufacturing employment. Second, across 275 US metropolitan statistical areas, drops in manufacturing employment among prime-age, non-college educated males are negatively correlated with demand for housing (measured as the sum of log changes in housing prices and approved building permits). Finally, evidence from a small sample of the Bureau of Labor Statistics' Displaced Worker Survey suggests that workers displaced from manufacturing were relatively more likely to transition to construction in areas with strong housing demand.

[17] Quasi-medians are based on means of groups of workers around the median, as Census Bureau disclosure avoidance procedures do not allow the reporting of true medians, which are necessarily based on one or two individuals. We caution that the estimates in Figure 1 contain a mix of voluntary and involuntary transitions, and that they may involve movement of select groups of workers. We condition on observed worker attributes in our regression analysis below.

Figure 1: Initial M Workers' Median 2000-7 Earnings Growth, by Destination Sector



Source: LEHD, LBD and authors' calculations. Figure displays the average 2007 versus 2000 earnings ratio across all workers moving from manufacturing to the noted 2-digit NAICS sector between 2000 and 2007 in the 46 states for which information is available in the LEHD for these years (Alabama, Arkansas, New Hampshire Mississippi and DC are excluded).

with a popular narrative that well-paid manufacturing workers face large drops in income when they move to low-skill service sectors (Scott et al., 2022). Workers transitioning to Wholesale (NAICS 42), by contrast, exhibit relatively large earnings growth. One explanation for the difference between Wholesale and Retail is the growing importance of increasingly sophisticated logistics skills required in the latter (Ding et al., 2019).

The ordering of sectors in Figure 1 also provide intuition for the transitions observed in Table 2. For example, some of the sectors that show up as the most popular size-adjusted destinations for workers in Table 2 – such as Mining and Wholesale – are among those with relatively large earnings gains in Figure 1. Similarly, some of the least-followed trajectories, such as towards Accommodation and Food, or Arts, Entertainment, and Recreation, are associated with substantial earnings declines. An exception is Education (NAICS 61), which is a relatively popular destination for former M workers despite yielding no gain in nominal earnings.

4 Industry and County Exposure to PNTR

The US granting of PNTR to China in October 2000 was unique in that it left assessed tariff rates unchanged, but altered the way US imports from China were considered under the two sets of tariffs that comprise the US Tariff Schedule. The first set of US tariffs, known as NTR tariffs, are applied to goods imported from fellow members of the World Trade Organization (WTO) and are generally, but not uniformly, low due to repeated rounds of trade negotiations during the post-war period. The second set of tariffs, known as non-NTR tariffs, were set by the Smoot-Hawley Tariff Act of 1930 and are often substantially higher than the corresponding NTR rates. Imports from non-market economies

such as China are by default subject to the higher non-NTR rates, but US law allows the President to grant such countries access to NTR rates on a year-by-year basis subject to annual approval by Congress.

US Presidents granted China such a waiver every year starting in 1980, but, as documented in Pierce and Schott (2016), Congressional votes over annual renewal became politically contentious and less certain of passage following various flash points in US-China relations, in particular the Chinese government's crackdown on Tiananmen Square protests in 1989. As a result, firms considering engaging in US-China trade prior to PNTR faced the possibility of substantial tariff increases, potenially making importer-exporter relationships unprofitable (Heise et al., 2022). This uncertainty ended with passage of PNTR, which "locked in" China's access to NTR tariff rates, eliminating the disincentive to US-China trade caused by the annual renewal process, and effectively liberalizing trade between the two countries.

Following Pierce and Schott (2016), we measure industry i's exposure to PNTR as the rise in US tariffs on Chinese goods that would have occurred in the event of a failed annual renewal of China's NTR status prior to PNTR's extension,

$$Industry\ Gap_i = Non\ NTR\ Rate_i - NTR\ Rate_i. \tag{1}$$

We compute $NTR\ Gap_i$ for six-digit NAICS industries using a simple average of the Harmonized system (HS) level *ad valorem equivalent* tariff rates provided by Feenstra et al. (2002) for the year 1999. The HS to NAICS concordance is developed by Pierce and Schott (2012). Our gap measure is only computed as of year 1999, which is the year before PNTR. As discussed in Pierce and Schott (2016), an attractive feature of this measure is its plausible exogeneity to employment outcomes after 2000, as 79 percent of the variation in the NTR gap across industries arises from variation in non-NTR rates, set 70 years before. This feature of non-NTR rates effectively rules out reverse causality that would arise if NTR rates were set to protect industries with declining employment or surging imports: to the extent such activity occurred, the higher NTR *rates* would result in a lower $Industry\ Gap_i$, biasing results away from finding an effect of the change in policy.

We follow Topalova (2007) and Pierce and Schott (2020) in computing a Bartik-style county exposure to PNTR as the employment-weighted average $NTR Gap_i$ of the industries it produces. For each US county c,

$$County\ Gap_c = \sum_i \frac{L_{ic}^{1990}}{L_c^{1990}} Industry\ Gap_i, \tag{2}$$

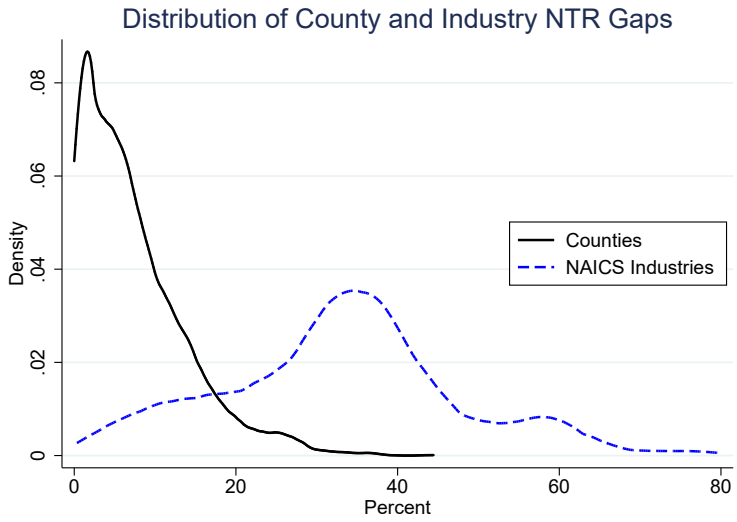
where the employment shares for 1990 are based on county-industry employment recorded in the US Census Bureau's Longitudinal Business Database (LBD), which tracks the employment of virtually all US firms and establishments from 1977 to the present.[18] In this computation, $Industry\ Gap_i$ is defined only for industries whose outputs are subject to US import tariffs, primarily in the manufac-

[18] An advantage of the LBD versus the more commonly used and publicly available County Business Patterns (CBP) for computing county-industry labor shares is, e.g., as in Autor et al. (2013) and Pierce and Schott (2020), is that it contains employment counts for all industries and counties, thereby avoiding issues of suppression to maintain confidentiality in the public version of the CBP (Eckert et al., 2020).

turing sector. For industries whose output is not subject to tariffs, such as service industries, we set $Industry\ Gap_i$ to zero.

Figure 2 displays the kernel densities of $Industry\ Gap_i$ and $County\ Gap_i$, where for ease of exposition, the former is restricted to industries that appear in the US tariff schedule. As a result, that distribution omits a large mass at zero representing non-goods industries that are not subject to tariffs. $Industry\ Gap_i$ has a mean and standard deviation of 33 and 14 percent, while $County\ Gap_i$ has a mean and standard deviation of 7 and 6 percent. Intuitively, the distribution of $County\ Gap_j$ lies to the left of the distribution of $Industry\ Gap_i$, reflecting the presence of service industries with NTR gaps of zero. The correlation between $Industry\ Gap$ and $County\ Gap$ across workers in our 19-state regression sample is 0.26.[19] When calculating the economic significance of estimated effects of PNTR in the regression analysis below, we make use of interquartile shifts in exposure, which are 24.6 and 6.9 percent for industry and county, respectively.[20] In Appendix Figure A.2, we report the average up- and downstream gap by two-digit NAICS sector.

Figure 2: Distribution of $Industry\ Gap$ and $County\ Gap$



Source: Longitudinal Business Database, Feenstra et al. (2002) and authors' calculations. Figure displays the distributions of the 1999 NTR gap across six-digit NAICS industries ($Industry\ Gap_i$) and US counties ($NTR\ Gap_c$). The former is restricted to the 473 industries that appear in the US tariff schedule. $Industry\ Gap$ has a mean and standard deviation of 28.4 and 17.4 percent, and an interquartile range of 24.6 percent. $County\ Gap_j$ has a mean and standard deviation of 6.1 and 5.2 percent, and an interquartile range of 6.9 percent.

Trade liberalization episodes such as PNTR may also affect US workers' earnings and employment indirectly via their supply chains, i.e., the upstream industries from which they purchase their inputs or the downstream industries to which they sell their outputs.[21] We compute up- and downstream

[19] Autor et al. (2014) report a correlation of 0.12 for their industry (four-digit SIC) and region (commuting zone) exposure to China, which are measured in terms of Chinese import penetration.

[20] The 10th, 25th, 50th, 75th and 90th percentiles of $Industry\ Gap_i$ are .05, .20, .33, .40 and .54. The corresponding percentiles for the $County\ Gap_j$ are 0.01, 0.02, 0.06, 0.10, and .15.

[21] A number of recent papers emphasize the importance of examining input-output linkages when estimating the impact of import competition, e.g., Amiti and Konings (2007); Goldberg et al. (2010); Acemoglu et al. (2016); Flaaen

NTR gaps using information from the 1997 BEA input-output tables. $Industry\,Gap_i^{up}$ is the weighted average of all 6-digit NAICS industries k used by industry i, using total-use input-output coefficients (ω_{ik}^{up}) as weights,

$$Industry\ \ Gap_i^{up} = \sum_k \omega_{ik}^{up}\ Industry\ \ Gap_k. \tag{3}$$

$Industry\ \ Gap_i^{down}$ is the analogous weighted average for all the downstream industries that use industry i. In constructing both of these measures, we exclude all k within the same 3-digit NAICS root of i.

We compute $County\ Gap_c^{up}$ and $County\ Gap_c^{down}$ by taking weighted averages of $Industry\ Gap_i^{up}$ and $Industry\ Gap_i^{down}$, e.g.,

$$County\ \ Gap_c^{up} = \sum_i \frac{L_{ic}^{1990}}{L_c^{1990}} Industry\ \ Gap_i^{up}. \tag{4}$$

Upstream exposure is therefore higher when the county produces more output in industries whose upstream industries have higher exposure.[22]

5 Earnings and Employment Responses

In this section, we examine the link between PNTR and worker earnings and employment using generalized OLS difference-in-differences (DID) specifications. This approach allows us to compare the impact of county versus industry exposure to the policy change while controlling for initial individual, firm, industry, and county characteristics, along with worker and time fixed effects, α_j and α_t. We examine whether the earnings and employment of workers with greater industry ($Industry\,Gap_i$) and county ($County\,Gap_c$) exposure to PNTR (first difference) vary after PNTR versus before (second difference),

$$\begin{aligned} d_{jfcit} \;=\; & \delta_1 Post \times County\,Gap_c + \delta_2 Post \times Industry\,Gap_i + \delta_3 Post \times MSH_{c,1999} + \\ & Post \times \mathbf{X_{j,1999}}\beta_{\mathbf{j}} + Post \times \mathbf{X_{f,1999}}\beta_{\mathbf{f}} + Post \times \mathbf{X_{it}}\beta_{\mathbf{i}} + \\ & \gamma_1 Post + \gamma_1 MSH_{c,1999} + + \mathbf{X_{it}}\gamma_{\mathbf{i}} + \alpha_j + \alpha_t + \epsilon_{jfcit}. \end{aligned} \tag{5}$$

All regressions are via ordinary least squares (OLS). The sample period is 1993 to 2014. The sample consists of a 5 percent stratified random draw from the 19 states for which employer-employee data are available in all these years. We exclude workers that are 65 years and older as of 2014 so as to exclude retirement. We weight observations by the inverse of the probability of being in the sample. The two dependent variables we consider are earnings and an indicator for employment. For earnings, we employ the inverse hyperbolic sine transform of the earnings of worker j in firm f in county c,

and Pierce (2019).

[22] The mean of $Industry\ Gap_i^{up}$, and $Industry\ Gap_i^{down}$, $County\ Gap_c^{up}$, and $County\ Gap_c^{down}$ are 4.8, 7.4, 5.3 and 6.3 percent. Their standard deviations are are 10.6, 6.8, 2.8 and 1.8 percent. Their interquartile ranges are 0.051, 0.066, 0.017 and .019 percent.

industry i in year t,

$$\widetilde{Earnings}_{jfcit} = arcsinh(Earnings_{jfcit}) = ln(Earnings_{jfcit} + \sqrt{Earnings_{jfcit}^2 + 1}), \tag{6}$$

which approximates log earnings and allows for zeros (Burbidge et al., 1988; Bellemare and Wichman, 2020). The implied elasticity of earnings with respect to (county-level) exposure is

$$\frac{\partial Earnings}{\partial County\ Gap_i}\frac{County\ Gap_i}{Earnings} = \delta_1 County\ Gap_i \sqrt{\frac{Earnings^2 + 1}{Earnings^2}} \approx \delta_1 County\ Gap_i, \tag{7}$$

so that the percent impact on earnings of an interquartile shift in county expsoure is approximately $100 \times \delta_1(County\ Gap_c^{75} - County\ Gap_c^{25})$, where, as noted earlier, the latter difference is 0.069. We measure employment as a dummy that takes the value 1 when the worker has paid LEHD employment in year t, $\mathbb{1}(Earnings_{jfcit} > 0)$. This measure is conservative in the sense that it does not account for reductions in the amount of employment within years.

Worker, firm, industry, and county attributes are defined as of the year prior to the policy change, 1999. The first two terms on the right-hand side of equation 5 are the county and industry DID terms of interest, i.e. interactions of county- or industry-level exposure to PNTR with a Post-PNTR dummy that takes the value 1 for years 2000 and later. The third term on the right-hand side represents county c's 1999 manufacturing share, $MSH_{c,1999}$. This term is a separate control for the prevalence of manufacturing in a county, thereby allowing the county exposure term to reflect exposure to PNTR conditional on the county's manufacturing share (Borusyak et al., 2021). The remaining terms on the right-hand side of equation 5 are controls for 1999 worker and firm characteristics interacted by the Post dummy, $Post \times \mathbf{X_{j,1999}}$ and $Post \times \mathbf{X_{f,1999}}$, and time-varying industry characteristics, $\mathbf{X_{it}}$.[23]

Initial worker attributes are age, gender, race, foreign-born status and education. Initial firm characteristics are firm-size categories, trading status, and diversification. Trading status is import only, export only, both or neither. Diversification is an indicator for whether or not the firm operates both manufacturing and non-manufacturing establishments. Industry characteristics capture other changes in policy that occur during our sample period: reductions in Chinese import tariffs, reductions in Chinese production subsidies, and the elimination of US quotas on textile and clothing products as part of the phasing out of the global Multifiber Arrangement (MFA). These variables are taken from (Pierce and Schott, 2016); their construction is described in Section B of the appendix.

We estimate equation 5 across four groups of workers: initial manufacturing (M) and non-manufacturing (NM) workers in each of our high- versus low-attachment samples. For ease of exposition, we refer to these four groups as $High\text{-}M$, $Low\text{-}M$, $High\text{-}NM$ and $Low\text{-}NM$. Table 3 summarizes the attributes of the workers in each of these groups as of the year 1999. As indicated in the table, initial manufacturing workers at this time are more likely to be male and have lower levels of education, and are somewhat less likely to be American-born. The attributes of high- and low-attachment workers are similar except that high-attachment workers are slightly older and have

[23] We need to multiply the 1999 worker and firm characteristics by the Post dummy since these characteristics do not change over time and the worker fixed effects would completely absorb them.

higher earnings.[24]

Table 3: 19-State Sample Worker Attributes in 1999

	High Attachment		Low Attachment	
	M	NM	M	NM
Female	0.29	0.47	0.30	0.48
	0.45	0.50	0.46	0.50
White	0.87	0.87	0.85	0.84
	0.34	0.34	0.36	0.37
American Born	0.85	0.89	0.85	0.89
	0.36	0.32	0.36	0.32
Less than HS	0.13	0.09	0.14	0.11
	0.33	0.28	0.35	0.31
HS	0.34	0.27	0.34	0.28
	0.47	0.44	0.47	0.45
Some College	0.32	0.34	0.33	0.34
	0.47	0.47	0.47	0.47
College or More	0.21	0.31	0.20	0.28
	0.41	0.46	0.40	0.45
Age	37.7	37.1	35.2	33.9
	6.4	6.7	7.4	7.8
Earnings	46,180	46,120	38,750	33,870
	226,070	217,100	89,400	113,200
Observations (000)	1,608	4,895	4,274	17,360

Source: LEHD, LBD and authors' calculations. Table reports the mean standard deviation of noted worker attributes in 1999 by whether workers are employed in manufacturing or not (M vs NM) and whether they are $High$ or Low attachment, where the former are defined as those employed by the same firm continuously from 1993 to 1999 and the latter as continuously employed from 1993 to 1999, but not necessarily by the same firm. Sample is restricted to a stratified random draw of the 19 states whose information is available in the LEHD from 1993 to 2014. Workers aged 64 and above in 2014 are excluded. The first six rows are shares. Age and earnings are in years and dollars.

5.1 Own-Exposure Specification (Baseline Specification)

In our baseline specification we follow the literature and restrict attention to own-county and own-industry exposure to PNTR. Table 4 reports our findings with respect to earnings using the specification in equation 5. The dependent variable is the inverse hyperbolic sine transform of the earnings of worker j in industry i and county c at firm f in year t, $\widetilde{Earnings}_{jfcit}$, as defined in equation (6). The four columns report the coefficients and standard errors for the DID terms of interest for all four groups of workers, noted at the bottom of the table. To conserve space, we do not report estimates

[24]For comparison, Appendix Table A.1 summarizes the attributes of all M and NM in the 46 states for which information is available in the LEHD in 2000. As indicated in the table, the M and NM workers in our 19-state regression are highly similar along the noted characteristics to those in the 46-state sample, with those in larger sample being a bit older, on average, than those in the 19-state sample.

for the other covariates included in the regression. Standard errors are two-way clustered by 4-digit NAICS and county.

Table 4: Baseline Earnings Results

	$\widetilde{Earnings}_{jfcit}$	$\widetilde{Earnings}_{jfcit}$	$\widetilde{Earnings}_{jfcit}$	$\widetilde{Earnings}_{jfcit}$
$Post_t$ x $Industry\ Gap_i$	0.125		-0.066	
	(0.190)		(0.116)	
$Post_t$ x $County\ Gap_c$	-3.612***	-3.578***	-3.174***	-5.176***
	(0.838)	(0.988)	(0.673)	(0.927)
R-Squared	0.453	0.456	0.445	0.446
Observations (000)	1,608	4,895	4,274	17,360
Fixed Effects	j,f,t	j,f,t	j,f,t	j,f,t
Two-way Clustering	N4,c	N4,c	N4,c	N4,c
Worker x *Post* Controls	Yes	Yes	Yes	Yes
Firm x *Post* Controls	Yes	Yes	Yes	Yes
Industry x *Post* Controls	Yes	Yes	Yes	Yes
Initial Attachment Sample	High	High	Low	Low
Initial Sector Sample	M	NM	M	NM
Interquartile ↑ in $County\ Gap_c$	-28.2%	-27.9%	-24.8%	-40.4%

Source: LEHD, LBD, and authors' calculations. Table displays DID terms of interest from estimation of equation 5 on the 1993 to 2014 earnings of worker j in year t, who was employed by firm f in county c in industry i in 1999. The four columns compare estimates for high- and low-attachment workers initially in manufacturing (M) and non-manufacturing (NM). $Post_t$ is a dummy variable for years after 2000. Gap_i and Gap_c are the industry and county exposures defined in Section 4. Regressions include noted controls and fixed effects. Standard errors two-way clustered by 4-digit NAICS and county are noted below coefficients. Regression samples are 5 percent stratified random draws from the 19 states whose data are available in the LEHD from 1993 to 2014. Workers aged 64 and above in 2014 are excluded. Observations are weighted by the inverse probability of being in the sample. Final row of the table reports the implied impact of an interquartile increase in county exposure in percentage terms. ***, **, and represent statistical significance at the 1, 5 and 10 percent levels.

We find that exposure to PNTR operates through workers' county rather than industry of employment. Coefficient estimates for industry exposure are close to zero and statistically insignificant for manufacturing workers, while those for county exposure are economically large and precisely estimated for all four groups of workers. The primacy of county exposure suggests workers face binding costs related to physical re-location in response to the shock.

The second major finding in Table 4 is that the trade shock spills over to workers initially employed outside manufacturing. This result suggests manufacturing and non-manufacturing workers compete in over-lapping labor markets within counties. The economic significance of our estimates is reported in the final row of the table, which reports the implied impact of an interquartile shift in county exposure on workers' earnings after the change in policy versus before. For high-attachment workers initially employed in manufacturing ($High\text{-}M$), such a shift reduces relative earnings by -28.2 percent. Impacts are similar for $High\text{-}NM$ and $Low\text{-}M$, but substantially more negative for $Low\text{-}NM$ workers, at -40.4 percent. One potential explanation of this result is that these workers may be most susceptible to competition from displaced M workers. Demand side forces may also play a role, in that this group

of younger, less-educated workers (see Table 3) may work disproportionately in service jobs such as retail or hospitality that are most sensitive to a decline in demand associated with the trade shock.

We assess the the contribution of lost employment on the estimated reductions in earnings reported above by estimating a linear probability version of equation 5 using as the outcome an indicator variable for whether we observe positive earnings for a worker in year t, $\mathbb{1}(Earnings_{jt} > 0)$. Results are reported analogously to earnings in Table 5. As indicated in the table, we find negative and statistically significant relationships between county exposure and post-period employment among all four groups of workers. Here, too, the effects are strongest among *Low-M* workers, where an interquartile increase in county exposure implies a relative drop in post-period employment of -3.1 percent. For the three other groups of workers, the analogous impacts are -1.8 to -2.2 percent. These results suggest that loss of employment accounts for a relatively small share of the overall decline in earnings noted in Table 4. For *High-M* workers, for example, the employment decline in response to an interquartile increase in county exposure is less than one tenth the loss in earnings (.076 = -2.0/-24.9). Shares are similar across the four groups of workers.[25]

Table 5: Baseline Employment Results

	$\mathbb{1}(Earnings_{jt} > 0)$	$\mathbb{1}(Earnings_{jt} > 0)$	$\mathbb{1}(Earnings_{jt} > 0)$	$\mathbb{1}(Earnings_{jt} > 0)$
Post x *Industry Gap*$_i$	0.009		-0.010	
	(0.015)		(0.009)	
Post x *County Gap*$_c$	-0.283***	-0.254***	-0.229***	-0.394***
	(0.070)	(0.079)	(0.059)	(0.079)
R-Squared	0.453	0.456	0.445	0.446
Observations (000)	1,608	4,895	4,274	17,360
Fixed Effects	j,f,t	j,f,t	j,f,t	j,f,t
Two-way Clustering	N4,c	N4,c	N4,c	N4,c
Worker x *Post* Controls	Yes	Yes	Yes	Yes
Firm x *Post* Controls	Yes	Yes	Yes	Yes
Industry x *Post* Controls	Yes	Yes	Yes	Yes
Initial Attachment Sample	High	High	Low	Low
Initial Sector Sample	M	NM	M	NM
Interquartile ↑ in *County Gap*$_c$	-2.2%	-2.0%	-1.8%	-3.1%

Source: LEHD, LBD, and authors' calculations. Table displays DID terms of interest from estimation of equation 5 on the 1993 to 2014 employment of worker j in year t, who was employed by firm f in county c in industry i in 1999. The four columns compare estimates for high- and low-attachment workers initially in manufacturing (M) and non-manufacturing (NM). $Post_t$ is a dummy variable for years after 2000. Gap_i and Gap_c are the industry and county exposures defined in Section 4. Regressions include noted controls and fixed effects. Standard errors two-way clustered by 4-digit NAICS and county are noted below coefficients. Regression samples are 5 percent stratified random draws from the 19 states whose data are available in the LEHD from 1993 to 2014. Workers aged 64 and above in 2014 are excluded. Observations are weighted by the inverse probability of being in the sample. Final row of table reports the impact of an interquartile shift in county exposure on post-period employment, in percentage terms. ***, **, and represent statistical significance at the 1, 5 and 10 percent levels.

The results in Tables 4 and 5 differ from existing research on US import competition. Autor et al. (2013), using commuting-zone-level data, find that greater exposure to imports from China has no

[25] One caveat that should be kept in mind when when evaluating these results is that $\mathbb{1}(Earnings_{jfcit} > 0)$ is a conservative measure of employment in that it evaluates to 1 if a worker j has *any* earnings in year t. As we do not observe hours worked, we are unable to determine the extent to which earnings decline due to fewer worked hours versus lower wage per hour.

impact on M wages but does lower NM wages, and that it reduces M employment but not NM employment. Examining worker-level data, Autor et al. (2014) and Hakobyan and McLaren (2016) find that regional and county exposure *both* are influential in US workers' response to import competition. In the former, an interquartile increase in regional exposure to Chinese import competition is associated with a 20 percent reduction in manufacturing workers' cumulative earnings and, at odds with our finding in Table 5, an increase in cumulative years employed. More recent commuting-zone-level research by Bloom et al. (2019), however, demonstrates that the M employment results in Autor et al. (2013) are sensitive to the time period examined, as well as which industrial classification system is used to trace employment, as manufacturing employment counts are different under SIC versus NAICS (Fort and Klimek, 2016). In their preferred specifications, Bloom et al. (2019) demontrate that greater regional exposure to China *raises* non-manufacturing employment. Though the coefficient estimates in Table 5 are at odds with that finding, results incorporating IO linkages reported in the next section are a closer fit.

We next consider whether county-level exposure affects all M workers in highly-exposed areas identically, or if the negative labor-market effects found above are concentrated among workers in particular industries. To do so, we augment equation 5 with a triple interaction, $Post$ x $Industry\ Gap_i$ x $County\ Gap_c$. (As industry exposure is not defined for NM workers, analysis is restricted to manufacturing workers.) Results are reported in Table 6. As shown in columns 1 and 2, for $High\text{-}M$, we continue to find that county-level exposure to PNTR is associated with lower cumulative earnings and a lower probability of employment, while the relationship with industry exposure remains statistically insignificant. In addition, the lack of statistical significance for the triple interaction indicates that for these $High\text{-}M$ workers, the effect of county exposure is present for all workers in the county, and is neither exacerbated nor dissipated by varying levels of industry exposure.

Among $Low\text{-}M$ workers, however, we find the negative implications of PNTR are concentrated among those workers who both live in a county that is highly exposed and are initially employed in a high-exposure industry. As shown in columns 3 and 4 of Table 6, for these workers, we find no relationship between outcomes and industry-level exposure, and a negative relationship with county-level exposure that is imprecisely estimated. However, we find a highly statistically significant relationship on the triple interaction term for both earnings and employment.[26] This concentration of the negative effects of the policy change among a smaller group of workers—particularly when they already had lower attachment to a current employer—provides additional explanation and context for why earlier research (Autor et al., 2019; Pierce and Schott, 2020) finds negative socioeconomic outcomes associated with higher import competition for some groups of workers.

[26]This result is similar to Costinot et al. (2022), who show more negative labor-market outcomes among workers at more highly exposed firms within more exposed regions in their analysis of Finnish workers' reactions to the implosion of the Soviet Union.

Table 6: Baseline Exposure Interactions

	$\widetilde{Earnings}_{jfcit}$	$\mathbb{1}(Earnings_{jt} > 0)$	$\widetilde{Earnings}_{jfcit}$	$\mathbb{1}(Earnings_{jt} > 0)$
Post x *Industry Gap*$_i$	0.13	0.005	0.276	0.015
	0.281	0.022	0.184	0.014
Post x *County Gap*$_c$	-3.587**	-0.300***	-1.458*	-0.1
	1.12	0.09	0.823	0.071
Post x *Industry Gap*$_i$ x *County Gap*$_c$	-0.071	0.048	-4.727***	-0.355***
	2.339	0.187	1.613	0.129
R-Squared	0.453	0.422	0.445	0.412
Observations (000)	1,608	1,608	4,274	4,274
Fixed Effects	j,t	j,t	j,t	j,t
Two-way Clustering	N4,c	N4,c	N4,c	N4,c
Worker x Post Controls	Yes	Yes	Yes	Yes
Firm x Post Controls	Yes	Yes	Yes	Yes
Industry x Post Controls	Yes	Yes	Yes	Yes
Initial Attachment Sample	High	High	Low	Low
Initial Sector Sample	M	M	M	M

Source: LEHD, LBD, and authors' calculations. Table displays DID and triple differences terms of interest from estimation of augmented version of equation 5 on the adjusted 1993 to 2014 earnings or an indicator for positive earnings of worker j in year t, who was employed by firm f in county c in industry i in 1999. The four columns compare estimates for high- and low-attachment workers initially in manufacturing (M). $Post_t$ is a dummy variable for years after 2000. Gap_i and Gap_c are the industry and county exposures defined in Section 4. Regressions include noted controls and fixed effects. Standard errors two-way clustered by 4-digit NAICS and county are noted below coefficients. Regression samples are 5 percent stratified random draws from the 19 states whose data are available in the LEHD from 1993 to 2014. Workers aged 64 and above in 2014 are excluded. Observations are weighted by the inverse probability of being in the sample. Final row of table reports the impact of an interquartile shift in county exposure on post-period employment, in percentage terms. ***, **, and represent statistical significance at the 1, 5 and 10 percent levels.

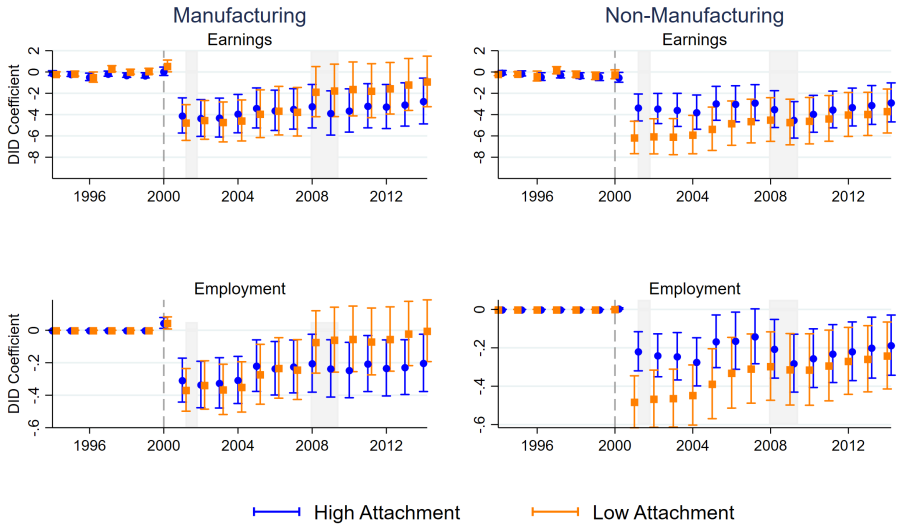
For the relative declines in earnings and employment in Tables 4 to 6 to be attributable to PNTR, our measure of exposure should be correlated with this outcome after PNTR but not before. To investigate whether this is so, we replace the $Post_t$ indicator in the first two terms of equation 5 with interactions of county and industry exposure and a full set of year dummies, omitting the first year, 1993. For both earnings and employment we find that terms with industry exposure are close to zero and not statistically significant at conventional levels, so we focus in Figure 3 on the county exposure interaction.[27]

Three trends stand out. First, for all four groups of workers, point estimates are near zero and statistically insignificant at conventional levels until 2001, at which time they drop substantially and become statistically significant. Thus, there are no pre-trends, and the impact of the shock is immediate. Second, the effect of the shock is long-lasting: for all but *Low-M* workers, earnings remain substantially below pre-PNTR levels through 2014. Third, the earnings of *Low-M* workers shift up in 2007, at the onset of the Great Recession. One potential explanation for this trend is that this recession, which coincided with another sharp drop in manufacturing employment, substantially worsened the earnings of relatively unexposed *Low-M* workers, bringing them closer to the more-

[27]We note that, by definition, $\mathbb{1}(Earnings_{jt} > 0) = 1$ in the pre-PNTR period for all workers in our samples.

exposed subset. Our finding of a persistent impact of PNTR on earnings is consistent with worker reactions to trade liberalization in Brazil (Dix-Carneiro and Kovak, 2017), the US (Autor et al., 2021; **?**), but differs from the dissipation of employment effects after 2007 in high-human-capital areas found in Bloom et al. (2019). Furthermore, our results indicate that the labor-market implications of exposure to PNTR are longer-lasting than those for Canadian workers associated with CUSFTA that are documented in Kovak and Morrow (2022).

Figure 3: County DID Coefficients from Annual Earnings Specification



Source: LEHD, LBD, and authors' calculations. Panels display the 95 percent confidence intervals for the estimated county exposure DID coefficients, i.e., interactions of year dummies with *County* Gap_c, for our four samples of workers: high- and low-attachment workers initially employed within and outside manufacturing. High attachment is defined as being continuously employed by the same firm in the pre period. Low attachment is defined as being continuously employed but not necessarily by the same firm over in the pre-period. Regressions include controls and fixed effects noted in the discussion around equation 5. Standard errors are two-way clustered by four-digit NAICS and county. Regression samples are 5 percent stratified random draws from the 19 states whose data are available in the LEHD from 1993 to 2014. Workers aged 64 and above in 2014 are excluded. Observations are weighted by the inverse probability of being in the sample. Number of observations is reported in Table 4. Shading corresponds to the 2001 and 2007 recessions.

5.2 IO Linkages Specification (Preferred Specification)

In this section we augment our baseline specification to consider workers' sensitivity to up- and downstream county and industry exposure using the measures constructed in Section 4. Upstream exposure is expected to raise worker earnings if greater openness with China results in lower input prices or otherwise affects productivity positively. Downstream exposure, by contrast, is expected to reduce worker earnings if it disrupts sales to customers. Including these additional covariates at the industry level is potentially most useful for NM workers, for whom direct industry exposure is not defined, e.g., designers using computers to sell engineering plans to automakers.

Results are reported in Table 7. We again find that exposure to the labor demand shock operates almost exclusively through workers' counties rather than industries. Here, however, there are substantial differences in the impact of county exposure among M and NM workers. First, while own-county

exposure continues to be negative for all four groups of workers, *vis à vis* the results above it is lower in magnitude among M workers, greater in magnitude for NM workers, and statistically significant only among the latter. Second, while county down- and upstream exposures have the expected negative and positive signs for all four groups of workers, the coefficients for upstream exposure are small in magnitude and imprecisely estimated among M workers, but large and statistically significant for NM workers. In terms of economic significance, interquartile shifts in $County\ Gap_c^{dn}$ are associated with relative declines in earnings ranging from -26.1 to -51.1 percent, while similar shifts in county upstream exposure induce relative earnings gains of 79.6 and 50.8 percent among $High\text{-}NM$ and $Low\text{-}NM$.[28]

[28] $Low\text{-}NM$ workers also exhibit lower earnings in response to downstream *industry* exposure. The estimated coefficient, -1.825, implies an earnings reduction of -12.0 percent ($= 100 \times -1.825 \times 0.066$) in response to an interquartile increase in industry exposure.

Table 7: Baseline Earnings Results with Up- and Downstream Exposure

	$\widetilde{Earnings}_{jfcit}$	$\widetilde{Earnings}_{jfcit}$	$\widetilde{Earnings}_{jfcit}$	$\widetilde{Earnings}_{jfcit}$
$Post_t$ x $Industry\ Gap_i$	0.141		-0.055	
	(0.201)		(0.125)	
$Post_t$ x $Industry\ Gap_i^{up}$	0.114	2.471	0.104	2.331
	(1.237)	(1.562)	(0.787)	(1.485)
$Post_t$ x $Industry\ Gap_i^{dn}$	-0.321	-1.187	-0.317	-1.825*
	(0.416)	(1.098)	(0.292)	(1.022)
$Post_t$ x $County\ Gap_c$	-2.154	-4.121***	-1.937	-5.417***
	(1.483)	(1.278)	(1.241)	(1.244)
$Post_t$ x $County\ Gap_c^{up}$	3.426	10.200**	1.937	6.507*
	(5.295)	(4.013)	(4.649)	(3.760)
$Post_t$ x $County\ Gap_c^{dn}$	-6.552***	-4.580***	-4.105**	-3.340**
	(2.211)	(1.663)	(1.618)	(1.583)
R-Squared	0.453	0.456	0.446	0.446
Observations (000)	1,608	4,895	4,274	17,360
Fixed Effects	j,f,t	j,f,t	j,f,t	j,f,t
Two-way Clustering	N4,c	N4,c	N4,c	N4,c
Worker x *Post* Controls	Yes	Yes	Yes	Yes
Firm x *Post* Controls	Yes	Yes	Yes	Yes
Industry x *Post* Controls	Yes	Yes	Yes	Yes
Initial Attachment Sample	High	High	Low	Low
Initial Sector Sample	M	NM	M	NM
Interquartile ↑ in $County\ Gap_c$	-16.8%	-32.1%	-15.1%	-42.3%
Interquartile ↑ in $County\ Gap_c^{up}$	26.7%	79.6%	15.1%	50.8%
Interquartile ↑ in $County\ Gap_c^{dn}$	-51.1%	-35.7%	-32.0%	-26.1%

Source: LEHD, LBD, and authors' calculations. Table displays DID terms of interest from estimation of equation 5 on the 1993 to 2014 earnings of worker j in year t, who was employed by firm f in county c in industry i in 1999. The four columns compare estimates for high- and low-attachment workers initially in manufacturing (M) and non-manufacturing (NM). $Post_t$ is a dummy variable for years after 2000. Gap_i and Gap_c are workers' industry and county exposures as defined in Section 4, with superscripts *up* and *down* refering to up- and downstream exposure via input-output linkages. Regressions include noted controls and fixed effects. Standard errors two-way clustered by 4-digit NAICS and county are noted below coefficients. Observations are weighted by the inverse probability of being in the sample. Regression samples are 5 percent stratified random draws from the 19 states whose data are available in the LEHD from 1993 to 2014. Workers aged 64 and above in 2014 are excluded. Final three row of table report the impact in percentage terms of an interquartile increase in noted county exposures. ***, **, and represent statistical significance at the 1, 5 and 10 percent levels.

Table 8, reporting analogous coefficient estimates for employment, reveals a largely similar pattern of results. The most substantial difference with respect to Table 7 is the insignificant role county upstream exposure plays for *Low-NM* workers, indicating that the upstream impact on *Low-M* earnings is relatlvely less dependent on employment than for *High-M* workers. The second major difference is the stronger influence of own-county exposure among *High-M* workers, suggesting a large role for employment in this group's earnings response.

Table 8: Baseline Employment Results with Up- and Downstream Exposure

	$\mathbb{1}(Earnings_{jt} > 0)$	$\mathbb{1}(Earnings_{jt} > 0)$	$\mathbb{1}(Earnings_{jt} > 0)$	$\mathbb{1}(Earnings_{jt} > 0)$
Post x *Industry Gap*$_i$	0.009		-0.011	
	(0.016)		(0.010)	
Post x *Industry Gap*$_i^{up}$	0.018	0.129	0.020	0.138
	(0.092)	(0.132)	(0.060)	(0.143)
Post x *Industry Gap*$_i^{dn}$	-0.015	-0.081	-0.007	-0.155
	(0.031)	(0.092)	(0.023)	(0.097)
Post x *County Gap*$_c$	-0.208*	-0.312***	-0.156	-0.425***
	(0.117)	(0.104)	(0.105)	(0.107)
Post x *County Gap*$_c^{up}$	0.368	0.776**	0.124	0.444
	(0.402)	(0.322)	(0.365)	(0.312)
Post x *County Gap*$_c^{dn}$	-0.460***	-0.290**	-0.268*	-0.187
	(0.174)	(0.134)	(0.143)	(0.141)
R-Squared	0.422	0.421	0.412	0.411
Observations (000)	1,608	4,895	4,274	17,360
Fixed Effects	j,f,t	j,f,t	j,f,t	j,f,t
Two-way Clustering	N4,c	N4,c	N4,c	N4,c
Worker x *Post* Controls	Yes	Yes	Yes	Yes
Firm x *Post* Controls	Yes	Yes	Yes	Yes
Industry x *Post* Controls	Yes	Yes	Yes	Yes
Initial Attachment Sample	High	High	Low	Low
Initial Sector Sample	M	NM	M	NM
Interquartile ↑ in *County Gap*$_c$	-1.6%	-2.4%	-1.2%	-3.3%
Interquartile ↑ in *County Gap*$_c^{up}$	2.9%	6.1%	1.0%	3.5%
Interquartile ↑ in *County Gap*$_c^{dn}$	-3.6%	-2.3%	-2.1%	-1.5%

Source: LEHD, LBD, and authors' calculations. Table displays DID terms of interest from estimation of equation 5 on the 1993 to 2014 employment of worker j in year t, who was employed by firm f in county c in industry i in 1999. The four columns compare estimates for high- and low-attachment workers initially in manufacturing (M) and non-manufacturing (NM). $Post_t$ is a dummy variable for years after 2000. Gap_i and Gap_c are workers' industry and county exposures as defined in Section 4, with superscripts *up* and *down* refering to up- and downstream exposure via input-output linkages. Regressions include noted controls and fixed effects. Standard errors two-way clustered by 4-digit NAICS and county are noted below coefficients. Observations are weighted by the inverse probability of being in the sample. Regression samples are 5 percent stratified random draws from the 19 states whose data are available in the LEHD from 1993 to 2014. Workers aged 64 and above in 2014 are excluded. Final three row of table report the impact in percentage terms of interquartile increases in noted county exposures. ***, **, and represent statistical significance at the 1, 5 and 10 percent levels.

The results in Tables 7 and 8 indicate that accounting for IO linkages is particularly important for understanding potential pro-competitive effects of trade liberalization. We quantify this importance at the county level by multiplying the county exposure coefficients from the baseline and preferred specifications by counties' actual exposures and comparing their distributions in Figure 4.[29] Predictions relying on the baseline specification are displayed using solid lines and are necessarily bounded from above by zero, as the baseline own-county exposure estimates are all negative. Distributions created using all three exposure variables from the preferred specification in Table 7 are dashed.[30]

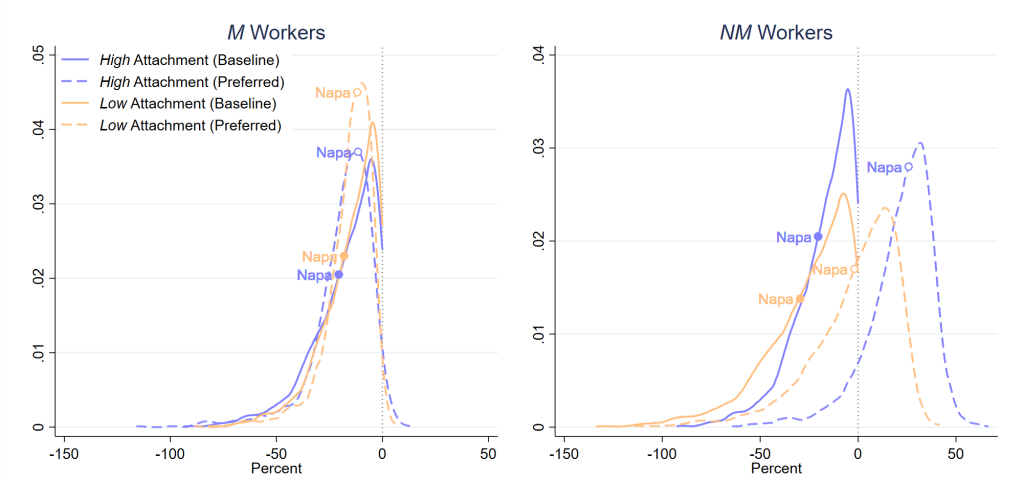
[29]Census disclosure guidelines preclude us from plotting analogous predictions at the worker level. We compare predictions of county-level relative earnings growth under the baseline versus preferred specifications to make an apples-to-apples comparison of their economic significance. Comparisons of interquartile increases in exposure are inadequate because increases in own, up- and downstream exposures cannot be done independently of one another.

[30]Comparison of predicted relative employment growth across counties using the baseline and IO-linkage specifications

As indicated in the figure, the two sets of predictions are similar for M workers (left panel) but very different for NM workers (right panel). Among the latter, predicted relative earnings growth using all three exposures lies substantially to the right of the distribution based exclusively on own exposure, particularly for $High-NM$ workers. An interesting case in point, highlighted in the figure, is Napa County, California. Napa has a relatively high upstream gap, lying at the 72nd percentile, versus its downstream gap, which sits at the 13th percentile. As a consequence, its predicted relative earnings growth jumps from a drop of -20.4 percent to an increase of 25.8 percent among $High$-NM workers, and from -29.6 to -0.1 percent among Low-NM workers. The latter is more anemic due to the relatively higher county upstream coefficient for $High$-NM workers. One potential explanation for this difference is that displaced M workers may be more of a substitute for Low-NM workers than $High$-NM workers.

Differences among M workers in Figure 4 are far less dramatic, both because of their relatively small county upstream exposure coefficients and their relatively large county downstream coefficients. For M workers, there are very few counties with positive predicted relative earnings growth using the estimates from Table 7.[31]

Figure 4: Distribution of Baseline vs Preferred Specification Predicted County Earnings Growth



Source: LEHD, LBD, and authors' calculations. Figure displays the distribution of predicted relative earnings growth across counties relying solely on estimates of own-county exposure from Table 4 (solid lines) versus estimates of own, up- and downstream exposure from Table 7 (dashed lines). Predictions for each county are obtained by multiplying the coefficients from the relevant table by $County\ Gap_c$, $County\ Gap_c^{up}$, and $County\ Gap_c^{dn}$. Vales for Napa County, CA are highlighted. Solid markers indicate Table 4 predictions while hollow markers indicate Table 7 predictions.

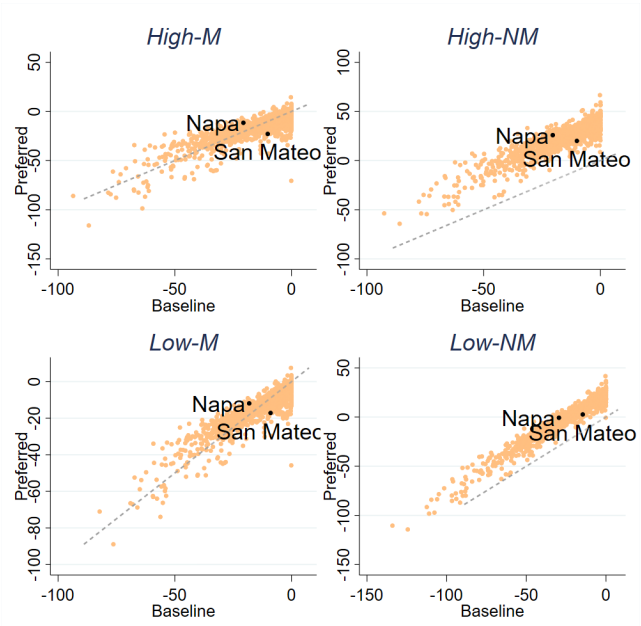
To examine the extent to which other counties are like Napa, Figure 5 plots predicted relative earnings growth for the baseline versus preferred specifications for all counties in our 19-state sample. Here, too, each point is a county. Points above the 45 degree line indicate greater predicted growth using the model with IO linkages, which is universally the case for NM workers. For M workers, however, counties lie both above and below the 45 degree line. As highlighted in the figure, San Mateo

yields figures that are very similar to Figures 4 and 5 for earnings. As a result, we omit them to conserve space.

[31] For $High$-NM workers predicted relative earnings are positive in 90 percent of counties representing 95 percent of workers. For Low-NM workers the analogous shares are 58 and 55 percent.

County, CA, otherwise known as Silicon Valley, serves as an interesting mirror-image of Napa in that it has comparatively low upstream exposure (29th percentile) but high downstream exposure (87th percentile). As a result, predicted relative earnings growth for San Mateo's M workers lies below the 45 degree line, i.e., it has *lower* predicted relative earnings growth when IO linkages are taken into account.

Figure 5: Predicted Baseline vs Preferred Specification County Earnings Growth



Source: LEHD, LBD, and authors' calculations. Figure displays the distribution of predicted relative earnings growth across counties relying solely on estimates of own-county exposure from Table 4 (solid lines) versus estimates of own, up- and downstream exposure from Table 7 (dashed lines). Predictions for each county are obtained by multiplying the coefficients from the relevant table by $County\ Gap_c$, $County\ Gap_c^{up}$, and $County\ Gap_c^{dn}$. Vales for Napa County, CA are highlighted. Solid markers indicate Table 4 predictions while hollow markers indicate Table 7 predictions.

In Appendix Figure A.3, we provide further intuition for the Napa-San Mateo comparison by reporting a breakdown of their initial employment shares, computed from an updated version of the County Business Patterns provided by Eckert et al. (2020). As shown in that figure, Napa is relatively specialized in Manufacturing (NAICS 3), Health (NAIC 62), and Accommodation and Food (NAICS 72), all of which are intensive in the use of inputs with higher gaps. San Mateo, by contrast, has relatively high employment shares in Transportation (NAICS 48-9), Professional Services (NAICS 54), Information (NAICS 51) and Wholesale (NAICS 42), all of which disproportionately serve higher-gap industries.[32]

The important role for upstream IO linkages found in this section is particularly noteworthy as existing research has found little evidence of a pro-competitive benefit of trade liberalization via this channel, e.g., Pierce and Schott (2016) and Acemoglu et al. (2016).[33] Results in Table 7, however,

[32]We report the average up- and downstream by 2-digit NAICS sector in Appendix Figure A.2.

[33]In other contexts – especially in developing countries – researchers have documented benefits to *firms* when input

indicate that the overall lack of a relationship in prior investigations masks important heterogeneity across levels of labor force attachment and, especially, sector.

6 Heterogeneous Outcomes

In this section, we examine whether responses to PNTR vary by worker characteristics using a version of equation 5 that includes interactions of workers' 1999 attributes with the county and industry exposure DID terms, e.g., *Post* x $Industry\ Gap_i \times Female$ and *Post* x $County\ Gap_c \times Female$. We run separate regressions for each group of workers and set of characteristics, i.e., four regressions for gender, another four for educational attainment, and so on. As triple interactions with the industry exposure DID term are not statistically significant at conventional levels, we concentrate our discussion on the triple interactions with the county exposure DID term.

Regressions coefficients are reported in Tables A.4 to A.7 of the Appendix. Here, we report their economic and statistical significance for earnings and employment visually in Figures 6 and 7.[34] There are four panels in each figure, one for each group of workers. The first bar in each panel repeats the implied impacts of interquartile shifts in county exposure from our baseline specification, i.e., the last rows of Tables 4 and 5. Each subsequent cluster of bars reports results for the noted dimension of worker heterogeneity, i.e., gender, race, and so on.[35] Bars are shaded black if the underlying coefficient estimate is statistically significant at the 10 percent level, and grey otherwise.

Worker Gender: A quick scan of the two figures reveals that our most consistent result is with respect to gender. We find positive relative earnings and employment responses for females in all four groups of workers, though the earnings results are statistically significant only for *High-NM* and *Low-NM* workers. The *positive* black bars in panels 2 and 4 of Figure 6 demonstrate that declines in earnings associated with an interquartile increase in county exposure to PNTR are 13.2 and 11.4 percentage points *less severe* for female *High-NM* and *Low-NM* workers than for corresponding males in those groups. By way of comparison, these magnitudes represent about one-half (13.2/-28.2) and one-fourth (11.4/-40.4) of the baseline implied impact of county exposure to PNTR (reproduced in the top row of the figure). The relative benefit to females with respect to employment is similar. The relatively better outcomes for females may reflect labor-market segmentation by gender, e.g., male *NM* wages being bid down by the entry of disproportionately male former *M* workers dislodged from manufacturing. Such an outcome would be consistent with the disproportionately negative effect of Chinese import competition on male socioeconomic outcomes documented in Autor et al. (2019) and Pierce and Schott (2020).

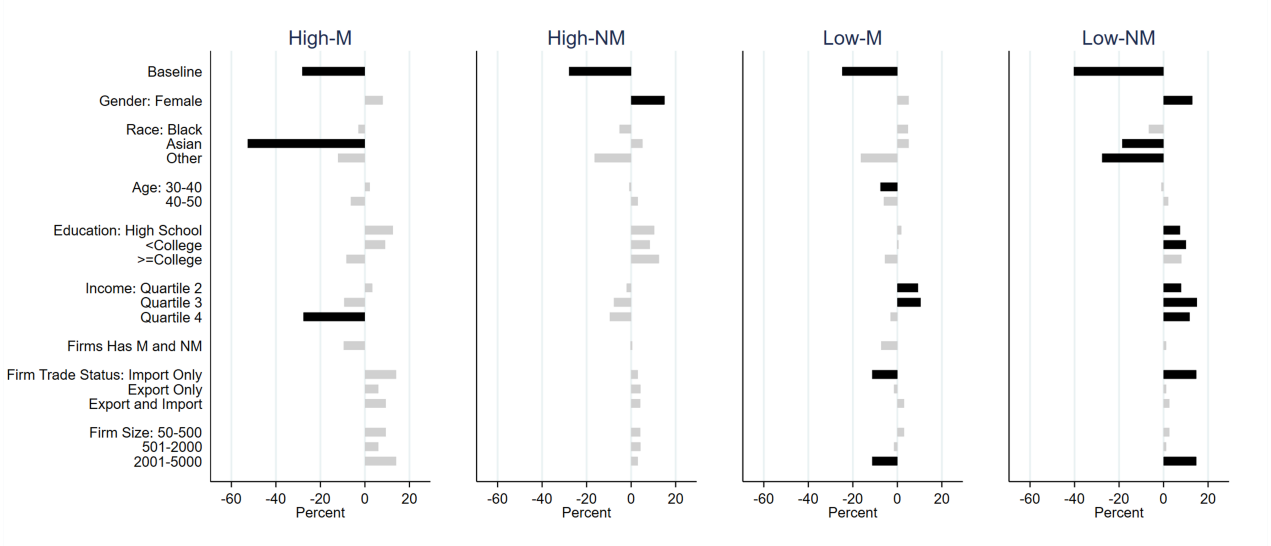
Worker Race: We find evidence that workers in some racial groups exhibit worsening in earnings and employment outcomes that exceeds that of whites, the left-out category. In particular, Asians

tariffs decline, including gains in productivity and product scope (Amiti and Konings, 2007; Goldberg et al., 2010).

[34]The interquartile inccreases in earnings and employment implied by the coefficients and plotted in Figures 6 and 7 are reported in Appendix Tables A.8 and A.9. For convenience, we assign the markers of statistical significance from the regression estimates in Appendix Tables A.4 to A.7 to their corresponding cells in Tables A.8 and A.9.

[35]The left out categories are: male for gender; white for race; below 30 for age, less than high school for educational attainment, the first (lowest) income quartile for income; worker's firm is only *M* or *NM* for diversification; neither imports nor exports for firm trading status, and firm has less than 50 employees for firm size.

Figure 6: Implied Impact on Earnings of Triple Interactions with County Exposure



Source: LEHD, LBD and authors' calculations. Figure displays the impact on earnings of an interquartile increase in county exposure implied by triple interactions of *Post* x *County Gap*$_c$ with worker or workers's firms' attributes. The first bar repeats the baseline impacts of from the final row of Table 4. Each subsequent set of bars is from a separate regression focusing on a different dimension of heterogeneity. Coefficients underlying the displayed impacts are reported in Appendix Tables A.4 and A.5; the actual magnitudes plotted appear in Appendix Table A.8. Black bars indicate that the underlying estimates are statistically significance at the 10 percent level or better.

exhibit starkly lower relative earnings drops among *High-M* and *Low-NM* workers, while a similar impact is observed among the "other" racial group in *Low-NM*. These patterns hold for employment, as well, with Asian employment also sharply affected among *H-NM* workers.

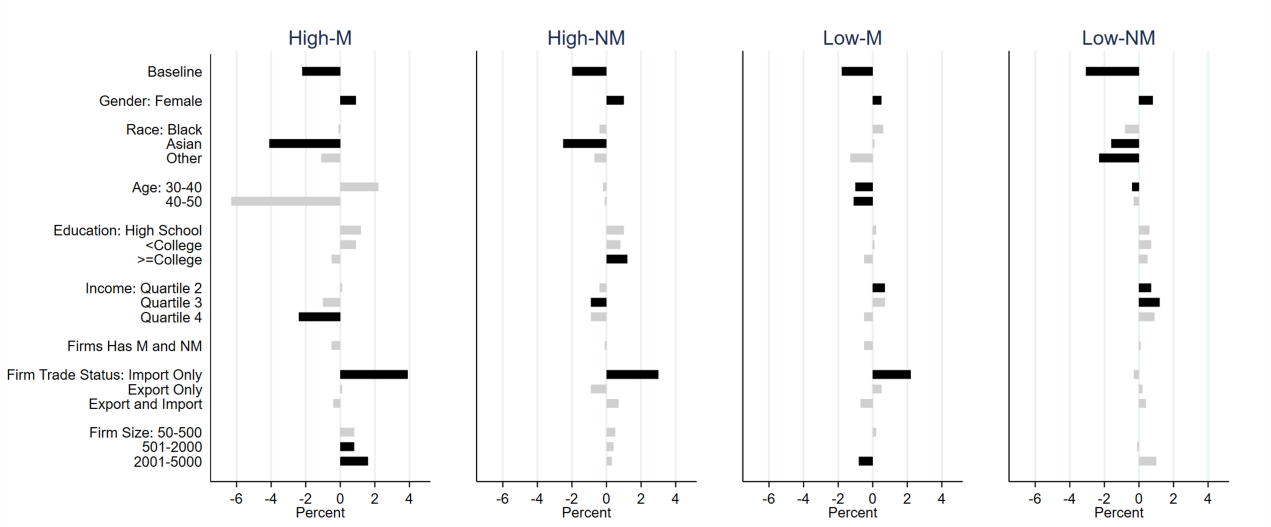
Initial Worker Age: Implied impacts with respect to wage are generally insignificant, though there is some some evidence that initially older workers fare worse than those initially under 30, the left-out category, among *Low-M* and to a lesser extent *Low-NM* workers.

Initial Worker Education: Implied impacts with respect to initial education are also largely insignificant, with some evidence that *Low-NM* workers with more education have relatively better earnings and employment outcomes.

Initial Worker Income: We examine workers' initial income quartiles as potentially revealing of human capital beyond formal education. For *Low* workers initially employed in either *M* or *NM*, we find that the relationship between exposure to PNTR and earnings is relatively better for workers whose initial incomes are above the first quartile. These results are consistent with similar findings for low-wage workers in Autor et al. (2014), and align with a story of displaced manufacturing workers searching for re-employment and pushing down wages for low-income, low-attachment workers.

Among *High-M* workers, we find that those in the highest initial income quartile experience declines in earnings due to county exposure to PNTR that are as large as the baseline impact (-27.6/-28.2), due in part to a higher probability of leaving employment (-2.4/-2.2). This striking pattern suggests that the highest-paid *M* workers may have firm- or *M* sector-specific human capital that is difficult to transfer to other sectors, leading to either reduced wages upon re-employment or a complete exit from the labor force, including via early retirement. This trend is consistent with the troubling finding in Sullivan and Wachter (2009) that high-seniority workers displaced from the

Figure 7: Implied Impact on Employment of Triple Interactions with County Exposure



Source: LEHD, LBD and authors' calculations. Figure displays the impact on earnings of an interquartile increase in county exposure implied by triple interactions of *Post* x *County Gap*$_c$ with worker or workers's firms' attributes. The first bar repeats the baseline impacts of from the final row of Table 5. Each subsequent set of bars is from a separate regression focusing on a different dimension of heterogeneity. Coefficients underlying the displayed impacts are reported in Appendix Tables A.6 and A.7; the actual magnitudes plotted appear in Appendix Table A.9. Black bars indicate that the underlying estimates are statistically significance at the 10 percent level or better.

collapsing steel industry in Pennsylvania in the 1980s experienced an elevated risk of early mortality.

Initial Firm Trading Status: One of the potentially most relevant firm characteristics for worker outcomes is initial trading status, i.e. whether the firm imports, exports, does neither, or both prior to PNTR. However, while we find that all M and $High\text{-}NM$ workers at firms that import only—versus those that don't trade (the excluded group), that only export, or that do both—exhibit relatively better employment outcomes, there does not seem to be an analogous impact on earnings. For $Low\text{-}NM$ workers we find the opposite: relatively better earnings with county exposure, but no impact on employment.

Initial Firm Size: One might expect firm size to be another important determinant of heterogeneity in workers' employment outcomes following PNTR. On the one hand, workers at larger firms may be more insulated from import competition if larger size is an indication of higher productivity or a higher ability to make fixed cost investments needed to retool production. On the other hand, smaller firms may face less increase in direct competition from imports if they are engaged in the production of customized or high-quality products (Holmes and Stevens, 2004). We find only minimal evidence for differences in responses for workers at firms of varying sizes. It is not an important dimension of heterogeneity for $High$ workers, and there also is little evidence for its relevance among Low workers, with only those at firms in the 2001 to 5000 employee range showing a difference from other firms. Furthermore, the differing signs for $Low\text{-}M$ and $Low\text{-}NM$ workers may simply indicate noise associated with one specific firm size class.

7 Conclusion

This paper provides a detailed analysis of how US workers responded to the precipitous decline in US manufacturing employment after 2000. Using linked employer-employee data from the US Census Bureau that have not previously been used to study the effects of the "China Shock", we begin by providing the first detailed accounting of workers' movements across sectors over this period, and earnings changes along each path. After accounting for the size of destination sectors, we show that wholesale, education, and mining (which includes oil and gas drilling and extraction) were the most popular destinations for former manufacturing workers, with accommodation and food services being among the least popular. The earnings implications of these moves provide some explanation for the observed movement between sectors, as the more popular destination sectors are often associated with earnings gains, while the least popular are associated with earnings losses. Specifically, we show that most workers who switch sectors experience nominal earnings gains by 2007, relative to their 2000 manufacturing earnings, with the largest gains accruing to workers who switch to mining, utilities, or management. However, workers who switch from manufacturing to accommodation or food services; arts, entertainment and recreation; or administrative and support and waste management experience outright nominal earnings declines.

Next, we use difference-in-differences analysis to examine US workers' earnings and employment after a 2000 change in US trade policy – PNTR – that effectively eliminated the possibility of a trade war. We consider both industry- and county-level measures of exposure to the policy change, examine workers initially employed in either manufacturing or non-manufacturing, and differentiate between effects for workers with low versus high attachment to their employers. Two results stand out. First, we find that exposure to this shock operates predominantly through workers' counties (versus industries), which likely reflects frictions in moving between different local labor markets, though this effect in high-exposure counties appears to be particularly acute for low-attachment manufacturing workers employed in high-exposure industries. Second, we show that accounting for workers' exposure along value chains is important for understanding the overall impact of the policy change, particularly for workers outside manufacturing. Indeed, while workers initially employed in manufacturing have substantial and persistent predicted declines in relative earnings, those outside manufacturing are generally predicted to experience relative earnings *gains*. Investigation of employment reactions indicates they account for a small share of the earnings effect.

Lastly, we examine heterogeneous responses to the policy change along a wide range of worker-, firm-, and industry-level dimensions, many of which have not been examined in the context of the US. We find that, in a relative sense, female workers' earnings hold up better than males' in response to PNTR, but that some Asian workers experience larger declines in earnings. Among low-attachment non-manufacturing workers—the group that experiences the most negative effects of PNTR—the largest earnings declines occur among those with less than a high school education and the lowest initial income. However, among high-attachment manufacturing workers, those in the *highest* initial income quartile experience *larger* earnings declines, indicating the presence of substantial fixed human capital associated with the initial firm or occupation of employment. Firm-level characteristics are generally less important for workers' responses to PNTR, though low-attachment non-manufacturing

workers fare relatively better if they are employed at a firm that imports but does not export or the largest category of firm.

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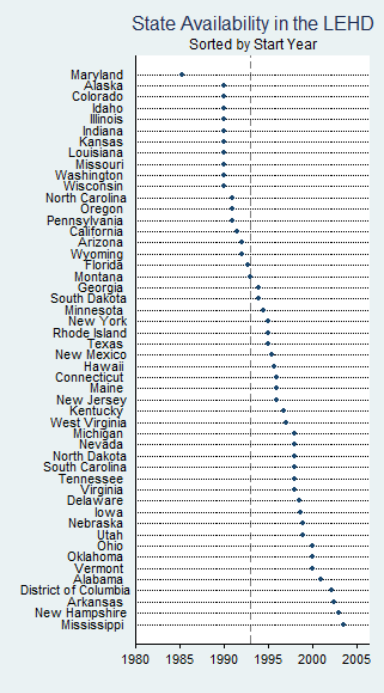
Online Appendix (Not for Publication)

This online appendix contains detailed additional empirical results as well as more detailed explanations of data used in the main text.

A State Coverage in the LEHD

The set of states included in the LEHD varies over time as summarized in Figure A.1. We use the 46 states available as of 2000 in examining worker movement between M and NM in Section 3, and the 19 states present from 1993 to 2014 for our regression analysis.

Figure A.1: State Availability in the LEHD



Source: Vilhuber and McKinney (2014). Figure displays the availability of state data in the LEHD.

B Industry Variable Construction

In this section we describe how the time-varying industry controls referenced in Section 5 are constructed.

MFA Exposure: As discussed in greater detail in Khandelwal et al. (2013), the MFA and its successor, the Agreement on Textile and Clothing (ATC), grew out of quotas imposed by the United States on textile and clothing imports from Japan during the 1950s. Over time, the MFA evolved into a broader institution that regulated the exports of clothing and textile products from developing countries to the United States, European Union, Canada and Turkey. Bargaining over these restrictions

was kept separate from multilateral trade negotiations until the conclusion of the Uruguay Round in 1995, when an agreement was struck to eliminate the quotas over four phases. On January 1, 1995, 1998, 2002 and 2005, the United States was required to remove textile and clothing quotas representing 16, 17, 18 and the remaining 49 percent of their 1990 import volumes, respectively. Relaxation of quotas on Chinese imports did not occur until it became a member of the World Trade Organization in 2001; as a result, its quotas on the goods in the first three phases were relaxed in early 2002 and its quotas on the goods in the fourth phase were relaxed as scheduled in 2005. The order in which goods were placed into a particular phase was chosen by the United States.

Computation of counties' exposure to elimination of the MFA proceeds in three steps. First, we follow Khandelwal et al. (2013) in measuring the extent to which MFA quotas in industry j and phase p were binding as the average fill rate of the industry's constituent import products in the year before they were phased out, $FillRate_{jp}$.[36] Specifically, for each phase, we measure an industry's exposure to MFA expiration as its average quota fill rate in the year prior to the phase's expiration. Industries with higher pre-expiration average fill rates faced more binding quotas and are therefore more exposed to the end of the MFA. Second, we compute counties' labor-share-weighted-average fill rate across industries for each phase, $FillRate_{cp}$. Finally, the county-year variable of interest, $MFAExposure_{ct}$, cumulates the calculated fill rates as each phase of expiration takes place. This measure of exposure to the MFA rises over time, as quotas for additional products are removed, by phase.

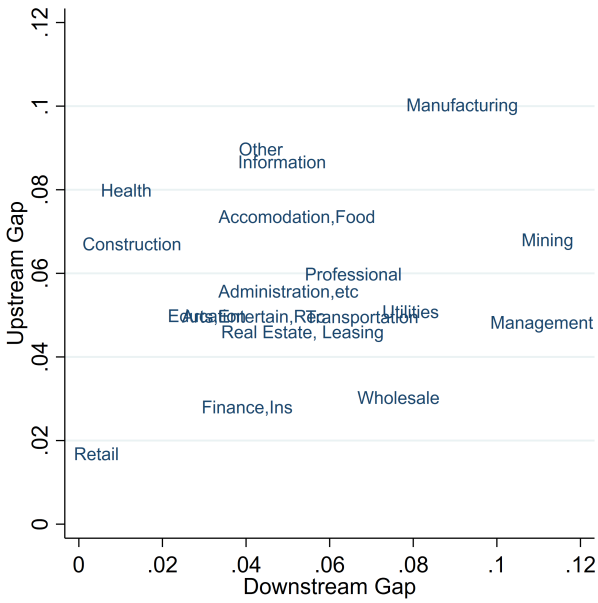
Changes in Chinese Policy: As part of its accession to the WTO, China agreed to institute a number of policy changes that could have influenced US manufacturing employment and thereby mortality, including liberalization of its import tariff rates and reductions of production subsidies, which might increase export opportunities for US manufacturers. Following Pierce and Schott (2016) we use product-level data on Chinese import tariffs from 1996 to 2005 from Brandt et al. (2017) to compute the average change across those years in Chinese import tariffs across products within each US industry. For production subsidies, we use data from the Annual Report of Industrial Enterprise Statistics compiled by China's National Bureau of Statistics (NBS), which reports the subsidies provided to responding firms.[37] For both changes in Chinese import tariff rates and production subsidies, we compute the labor-share-weighted average of this change across the industries active in each US county and then interact these variables with an indicator for post-PNTR years.

Up- and Downstream NTR Gaps: Figure A.2 reports the average up- and downstream NTR gaps across 6-digit NAICS industries within 2-digit NAICS sectors. Figure A.3 reports a breakdown of initial employment shares for Napa County and San Mateo County, both in California.

[36] As discussed in Brambilla et al. (2010), fill rates are defined as actual imports divided by allowable imports under the the quota. MFA products for which there were no restrictions on imports (i.e., there were no quotas), have fill rates of zero.

[37] The NBS data encompass a census of state-owned enterprises (SOEs) and a survey of all non-SOEs with annual sales above 5 million Renminbi (˜$600,000). The version of the NBS dataset available to us from Khandelwal et al. (2013) spans the period 1998 to 2005. Following Girma et al. (2009) and Aghion et al. (2015) we use the variable "subsidy" in this dataset and compute the change in the subsidies to sales ratio for each SIC industry between 1998 and 2005 using concordances provided by Dean and Lovely (2010).

Figure A.2: Average Up- and Downstream Gaps Across 2-Digit NAICS Sectors



Source: CBP, BEA, Feenstra et al. (2002) and authors' calculations. Figure displays mean up- and downstream NTR gap across 2-digit NAICS sectors. See Table 2 for sector labels.

C Worker Characteristics in the 46-State Sample

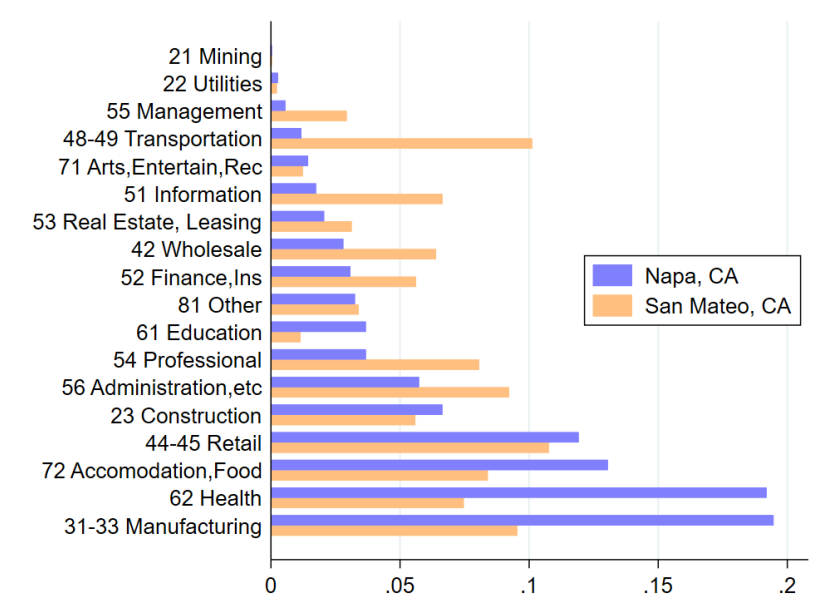
Table A.1 reports M and NM worker characteristics in 2000 across the 46 states for which information is available in the LEHD in that year.

Table A.1: US Worker Characteristics

	2000			
	Manufacturing		Non-Manufacturing	
	Mean	SD	Mean	SD
Male	0.67	0.47	0.49	0.50
American Born	0.83	0.38	0.87	0.34
≤High School	0.16	0.37	0.14	0.34
=High School	0.33	0.47	0.28	0.45
Some College	0.31	0.47	0.31	0.46
≥College	0.20	0.40	0.28	0.45
Age	39.7	12.86	37.3	14.51
Earnings	36,000	200,000	27,000	130,000

Source: LEHD, LBD and authors' calculations. Table reports the mean and standard deviation of noted worker attributes in 2000. Right and left panels compare workers initially employed in manufacturing to those initially employed outside manufacturing. Data are restricted to the 46 states whose information is available in the LEHD in 2000. All figures are in percent except age and earnings, which are in years and dollars.

Figure A.3: Napa versus San Mateo Employment Shares



Source: Eckert et al. (2020) (CBP) and authors' calculations. Figure displays the 1993 employment shares by 2-digit NAICS for noted counties. Sectors are sorted according to Napa's shares.

D Transitions for Other Time Periods

Table A.2 reports manufacturing to non-manufacturing transitions for two alternate time periods, 2000 to 2005 and 2000 to 2007, compared to the results reported in Table 1 in the main text, for 2000 to 2007. As in Table 1, the top and bottom panels report transitions in terms of millions of workers and percentages of initial levels, respectively. As indicated in the figure, gross flows out of initial sectors are lower for 2000 to 2005 than for 2000 to 2007, but substantially higher for 2000 to 2011 due to the intervening Great Recession.

Table A.2: 2000-5 and 2000-11 $M \leftrightarrow NM$ Transitions (46-State Sample)

	2000-2005				2000-2011			
	Employment (Millions)				Employment (Millions)			
Origin/Destination	NM	M	NE	Total	NM	M	NE	Total
Not Manufacturing (NM)	89.8	3.6	25.8	119.2	73.7	3.4	40.5	117.6
Manufacturing (M)	5.3	8.9	3.4	17.7	5.8	5.8	6.0	17.5
Not Employed (NE)	34.5	2.7	.	37.1	24.4	2.1	.	26.5
Total	129.5	15.2	29.3	174.0	103.8	11.2	46.5	161.5
	Share of Initial Level (Percent)				Share of Initial Level (Percent)			
Origin/Destination	NM	M	NE	Total	NM	M	NE	Total
Not Manufacturing (NM)	75	3	22	100	62	3	34	99
Manufacturing (M)	30	51	19	100	33	33	34	99
Not Employed (NE)	93	7	.	100	66	6	.	71
Total	74	9	17	100	60	6	27	93

Source: LEHD, LBD and authors' calculations. Table reports the transition paths of employed and not-employed workers from 2000 (row, left panel) to 2005 (column, left panel), and from 2000 (row, right panel) to 2011 (column, right pane), for the 46 states whose information is available in the LEHD starting in 2000. Alabama, Arkansas, New Hampshire and Mississippi as well as the District of Columbia are excluded. Upper panel reports levels in millions of workers. Lower panel reports shares of initial levels.

Table A.2 and Figure A.4 provides a more detailed version of the left panel of Table A.2 by reporting beginning and ending employment at the 1-digit NAICS sector level. As indicated in the table, the largest net losses of manufacturing employment are due to Not Employed (-.70 million), Business Services (-.6 million), Wholesale, Retail, Transportation and Warehousing (.5 million), Education and Health (-.42 million), and Mining, Utilities, and Construction (-.22 million). Only two 1-digit sectors, Agriculture, Forestry, Fishing and Hunting, and Arts, Entertainment, Accomodation and Food exhibit net inflows into manufacturing, of .04 and .05 million, respectively.

Table A.3: 2000-5 Transitions by 1-digit NAICS Sector (46-State Sample)

Employment (Millions)

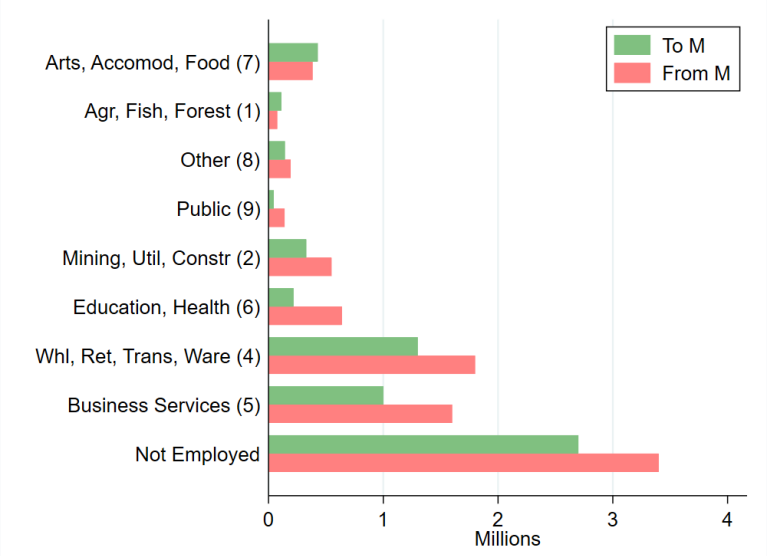
Sector in 2000	Sector in 2005										Total in
	1	2	3	4	5	6	7	8	9	NE	2000
Agriculture, Forestry, Fishing and Hunting(1)	0.57	0.09	0.11	0.13	0.11	0.05	0.06	0.02	0.01	0.62	1.79
Mining, Utilities, Construction (2)	0.04	5.43	0.33	0.59	0.83	0.30	0.21	0.11	0.18	2.49	10.49
Manufacturing (3)	0.08	0.55	8.94	1.78	1.55	0.64	0.38	0.19	0.14	3.43	17.68
Wholesale, Retail, Transportation, Warehousing (4)	0.09	0.88	1.29	13.83	3.22	1.78	1.19	0.50	0.34	6.08	29.18
Business Services (5)	0.07	0.93	1.04	2.43	14.79	1.99	0.97	0.41	0.39	6.70	29.73
Education, Healthcare (6)	0.02	0.22	0.22	0.85	1.41	16.00	0.44	0.25	0.38	4.35	24.14
Arts, Entertainment, Accomodation, Recreation (7)	0.04	0.42	0.43	1.61	1.67	1.16	4.73	0.26	0.16	3.36	13.84
Other Services (except Public Administration) (8)	0.01	0.13	0.14	0.43	0.43	0.36	0.18	1.52	0.06	1.27	4.52
Public Administration (9)	0.01	0.12	0.05	0.18	0.28	0.38	0.09	0.04	3.41	0.95	5.51
Not Employed (NE)	0.75	2.80	2.65	8.50	7.25	5.54	7.23	1.50	0.89		37.11
Total in 2005	1.69	11.56	15.20	30.33	31.53	28.20	15.49	4.81	5.94	29.25	174.00

Employment as a Percent of Initial Level

Sector in 2000	Sector in 2005										Total in
	1	2	3	4	5	6	7	8	9	NE	2000
Agriculture, Forestry, Fishing and Hunting(1)	32	5	6	7	6	3	4	1	1	35	100
Mining, Utilities, Construction (2)	0	52	3	6	8	3	2	1	2	24	100
Manufacturing (3)	0	3	51	10	9	4	2	1	1	19	100
Wholesale, Retail, Transportation, Warehousing (4)	0	3	4	47	11	6	4	2	1	21	100
Business Services (5)	0	3	3	8	50	7	3	1	1	23	100
Education, Healthcare (6)	0	1	1	4	6	66	2	1	2	18	100
Arts, Entertainment, Accomodation, Recreation (7)	0	3	3	12	12	8	34	2	1	24	100
Other Services (except Public Administration) (8)	0	3	3	9	9	8	4	34	1	28	100
Public Administration (9)	0	2	1	3	5	7	2	1	62	17	100
Not Employed (NE)	2	8	7	23	20	15	19	4	2		100
Total in 2005	1	7	9	17	18	16	9	3	3	17	100

Source: LEHD, LBD and authors' calculations. Table reports the transition paths of employed and not-employed workers from 2000 (row, left panel) to 2005 (column, left panel) by 1-digit NAICS sector (in parentheses) for the 46 states whose information is available in the LEHD starting in 2000. Alabama, Arkansas, New Hampshire and Mississippi as well as the District of Columbia are excluded. Upper panel reports levels in millions of workers. Lower panel reports shares of initial levels.

Figure A.4: Gross Flows Into and Out of M, 2000-5



Source: LEHD, LBD and authors' calculations. Figure displays the grows inflows into and gross outflows out of manufacturing between 2000 and 2005, in millions, by 1-digit NAICS sector (noted in parentheses) other than the diagonal. The full transition matrix (including the diagonal) is reported in Appendix Table A.3. Secrtors are sorted by net flows: .05, .04, -.05, -.09, -.22, -.42, -.5, -.6, and -.7.

E Heterogeneous Impacts

Tables A.4 to A.7 report the results of interacting the county and industry DID terms in our baseline specification (equation 5 with a series of worker attributes, defined as of 1999, e.g., $Post \times County\ Gap \times Gender$. As coefficients on the industry exposure triple interactions are not statistically significant at conventional levels, we conserve space by reporting coefficients and standard errors solely for the county exposure triple-interactions. Tables A.4 and A.5 report results for earnings, while Tables A.6 and A.7 summarize findings for employment. Each column of each panel reports these coefficients and their standard errors for a separate regression which includes only the noted attributes. Tables A.8 and A.9 report the impacts of interquartile increases in county exposure implied by the estimates in Tables A.4 and A.5, and displayed in Figures 6 and 7 of the main text.

Table A.4: Worker-Attribute Triple Interactions with County Exposure DID Term in Earnings Regressions

	$\widetilde{Earnings}_{jfcit}$	$\widetilde{Earnings}_{jfcit}$	$\widetilde{Earnings}_{jfcit}$	$\widetilde{Earnings}_{jfcit}$
Sample	*High-M*	*High-NM*	*Low-M*	*Low-NM*
		Gender		
Female	1.024	1.919***	0.655	1.651***
	(0.822)	(0.531)	(0.408)	(0.546)
R-Squared	.45	.46	.45	.45
		Race		
Black	-0.372	-0.672	0.619	-0.852
	(1.170)	(0.596)	(0.596)	-0.837
Asian	-6.761**	0.657	0.657	-2.386**
	(2.700)	(1.134)	(1.134)	-1.163
Other	-1.551	-2.102	-2.102	-3.541**
	(2.085)	(1.391)	(1.391)	-1.683
R-Squared	.45	.46	.45	.45
		Age		
30-40	0.276	-0.098	-0.978**	-0.132
	(0.878)	(0.599)	(0.471)	-0.343
40-50	-0.806	0.381	-0.781	0.265
	(0.978)	(0.734)	(0.728)	-0.477
R-Squared	.45	.46	.45	.45
		Education		
High	1.613	1.338	0.230	0.953*
	(1.231)	(1.031)	(0.520)	(0.550)
<College	1.169	1.073	0.050	1.299*
	(1.117)	(1.015)	(0.522)	(0.676)
≥College	-1.064	1.604	-0.723	1.023
	(1.575)	(1.098)	(0.844)	(0.941)
R-Squared	.45	.46	.45	.45
		Income		
Quartile 2	0.428	-0.255	1.187***	1.014***
	(0.641)	(0.662)	(0.420)	(0.371)
Quartile 3	-1.189	-0.992	1.352*	1.928***
	(1.008)	(0.650)	(0.689)	(0.566)
Quartile 4	-3.538**	-1.231	-0.396	1.494*
	(1.695)	(0.904)	(1.411)	(0.890)
R-Squared	.46	.46	.45	.45

Source: LEHD, LBD and authors' calculations. Each column of each panel of the table displays results from regressions analogous to those reported in Table 4 but including triple-interactions of the noted attributes with both county and industry DID terms of interest. Only the former are included here. ***, **, and represent statistical significance at the 1, 5 and 10 percent levels.

Table A.5: Firm-Attribute Triple Interactions with County Exposure DID Term in Earnings Regressions

	$\widetilde{Earnings}_{jfcit}$	$\widetilde{Earnings}_{jfcit}$	$\widetilde{Earnings}_{jfcit}$	$\widetilde{Earnings}_{jfcit}$
Sample	*High-M*	*High-NM*	*Low-M*	*Low-NM*
		Firm Diversification		
Firm has M and NM	-1.212	0.026	-0.931	0.153
	(0.750)	(0.976)	(0.713)	(0.651)
R-Squared	.45	.46	.45	.45
		Firm Trading Status		
Export and Import	-1.030	1.210*	-1.240	0.627
	(1.000)	(0.676)	(0.749)	(0.726)
Export Only	0.097	-1.601*	0.651	0.061
	(1.109)	(0.869)	(0.464)	(0.680)
Import Only	5.043*	5.099**	2.903**	-0.130
	(2.962)	(2.324)	(1.293)	(1.359)
R-Squared	.45	.46	.45	.45
		Firm Size (Employment)		
50-500	1.208	0.532	0.381	0.336
	(0.917)	(0.841)	(0.670)	(0.490)
501-2000	0.769	0.542	-0.191	0.152
	(0.716)	(1.003)	(0.616)	(0.964)
2001-5000	1.797	0.388	-1.444**	1.885**
	(1.186)	(1.519)	(0.573)	(0.818)
>5000	-1.607	0.971	-1.358	0.818
	(1.175)	(0.667)	(1.122)	(0.736)
R-Squared	.45	.46	.45	.45

Source: LEHD, LBD and authors' calculations. Each column of each panel of the table displays results from regressions analogous to those reported in Table 4 but including triple-interactions of the noted attributes with both county and industry DID terms of interest. Only the former are included here. ***, **, and represent statistical significance at the 1, 5 and 10 percent levels.

Table A.6: Worker-Attribute Triple Interactions with County Exposure DID Term in Employment Regressions

	$\mathbb{1}(Earnings_{jt} > 0)$	$\mathbb{1}(Earnings_{jt} > 0)$	$\mathbb{1}(Earnings_{jt} > 0)$	$\mathbb{1}(Earnings_{jt} > 0)$
Sample	*High-M*	*High-NM*	*Low-M*	*Low-NM*
		Gender		
Female	0.115*	0.134***	0.069*	0.099**
	(0.067)	(0.046)	(0.035)	(0.042)
R-Squared	.42	.42	.41	.41
		Race		
Black	-0.014	-0.057	0.082	-0.104
	(0.102)	(0.107)	(0.053)	-0.081
Asian	-0.525**	-0.323**	0.013	-0.208*
	(0.216)	(0.147)	(0.110)	-0.109
Other	-0.137	-0.096	-0.161	-0.293*
	(0.170)	(0.178)	(0.115)	-0.154
R-Squared	.42	.42	.41	.41
		Age		
30-40	0.276	-0.031	-0.123***	-0.054*
	(0.878)	(0.055)	(0.041)	-0.031
40-50	-0.806	-0.017	-0.138**	-0.035
	(0.978)	(0.064)	(0.061)	-0.037
R-Squared	.42	.42	.41	.41
		Education		
High	0.151	0.129	0.023	0.071
	(0.113)	(0.092)	(0.045)	(0.053)
<College	0.117	0.102	0.011	0.096
	(0.103)	(0.089)	(0.046)	(0.061)
≥College	-0.060	0.156*	-0.061	0.069
	(0.139)	(0.093)	(0.067)	(0.079)
R-Squared	.42	.42	.41	.41
		Income		
Quartile	0.011	-0.051	0.086**	0.092***
	(0.056)	(0.060)	(0.037)	(0.034)
Quartile	-0.128	-0.112*	0.093	0.159***
	(0.084)	(0.058)	(0.057)	(0.050)
Quartile	-0.308**	-0.117	-0.066	0.120
	(0.131)	(0.079)	(0.111)	(0.073)
R-Squared	.43	.42	.41	.42

Source: LEHD, LBD and authors' calculations. Each column of each panel of the table displays results from regressions analogous to those reported in Table 5 but including triple-interactions of the noted attributes with both county and industry DID terms of interest. Only the former are included here. ***, **, and represent statistical significance at the 1, 5 and 10 percent levels.

Table A.7: Firm-Attribute Triple Interactions with County Exposure DID Term in Employment Regressions

	$\mathbb{1}(Earnings_{jt} > 0)$	$\mathbb{1}(Earnings_{jt} > 0)$	$\mathbb{1}(Earnings_{jt} > 0)$	$\mathbb{1}(Earnings_{jt} > 0)$
Sample	*High-M*	*High-NM*	*Low-M*	*Low-NM*
		Firm Diversification		
Firm has M and NM	-0.067	-0.014	-0.061	0.016
	(0.061)	(0.080)	(0.059)	(0.051)
R-Squared	.42	.42	.41	.41
		Firm Trading Status		
Export and Import	-0.047	0.087	-0.085	0.051
	(0.078)	(0.057)	(0.064)	(0.053)
Export Only	0.014	-0.120	0.065	0.026
	(0.094)	(0.079)	(0.045)	(0.055)
Import Only	0.503*	0.379**	0.287**	-0.036
	(0.262)	(0.177)	(0.113)	(0.123)
R-Squared	.42	.42	.41	.41
		Firm Size (Employment)		
50-500	0.100	0.058	0.023	0.006
	(0.075)	(0.068)	(0.059)	(0.046)
501-2000	0.099*	0.055	-0.003	-0.010
	(0.056)	(0.089)	(0.053)	(0.085)
2001-5000	0.201*	0.037	-0.102**	0.131
	(0.103)	(0.126)	(0.049)	(0.079)
>5000	-0.083	0.057	-0.092	0.046
	(0.089)	(0.057)	(0.094)	(0.058)
R-Squared	.42	.42	.41	.41

Source: LEHD, LBD and authors' calculations. Each column of each panel of the table displays results from regressions analogous to those reported in Table 5 but including triple-interactions of the noted attributes with both county and industry DID terms of interest. Only the former are included here. ***, **, and represent statistical significance at the 1, 5 and 10 percent levels.

Table A.8: Impact on Earnings of Interquartile ↑ in *County Gap*$_c$ from Triple Interactions

	$\widetilde{Earnings}_{jfcit}$	$\widetilde{Earnings}_{jfcit}$	$\widetilde{Earnings}_{jfcit}$	$\widetilde{Earnings}_{jfcit}$
Sample	*High-M*	*High-NM*	*Low-M*	*Low-NM*
Baseline Impact	-28.2%	-27.9%	-24.8%	-40.4%
		Gender		
Female	8.0%	15.0%***	5.1%	12.9%***
		Race		
Black	-2.9%	-5.2%	4.8%	-6.6%
Asian	-52.7%**	5.1%	5.1%	-18.6%**
Other	-12.1%	-16.4%	-16.4%	-27.6%**
		Age		
30-40	2.2%	-0.8%	-7.6%**	-1.0%
40-50	-6.3%	3.0%	-6.1%	2.1%
		Education		
High	12.6%	10.4%	1.8%	7.4%*
<College	9.1%	8.4%	0.4%	10.1%*
≥College	-8.3%	12.5%	-5.6%	8.0%
		Income		
Quartile 2	3.3%	-2.0%	9.3%***	7.9%***
Quartile 3	-9.3%	-7.7%	10.5%*	15.0%***
Quartile 4	-27.6%**	-9.6%	-3.1%	11.7%*
		Firm Diversification		
Firm	-9.5%	0.2%	-7.3%	1.2%
		Firm Trading Status		
Export and Import	9.4%	4.1%	3.0%	2.6%
Export Only	6.0%	4.2%	-1.5%	1.2%
Import Only	14.0%	3.0%	-11.3%**	14.7%**
		Firm Size (Employment)		
50-500	9.4%	4.1%	3.0%	2.6%
501-2000	6.0%	4.2%	-1.5%	1.2%
2001-5000	14.0%	3.0%	-11.3%**	14.7%**
¿5000	-12.5%	7.6%	-10.6%	6.4%

Source: LEHD, LBD and authors' calculations. Each column of each panel of the table displays the implied impact on earnings of an interquartile increase in county exposure implied by a triple interaction of *Post* x *County Gap*$_c$ with noted row dummy variables capturing worker attributes. Coefficients are reported in Tables A.4 and A.5. First two rows note the sample and the baseline impact of an interquartile increase in county exposure from the final row of Table 4. ***, **, and represent statistical significance at the 1, 5 and 10 percent levels of the underlying triple DID coefficients.

Table A.9: Implied % Impact on Employment of Interquartile Increase in Exposure from Triple Interactions

	$\mathbb{1}(Earnings_{jt} > 0)$	$\mathbb{1}(Earnings_{jt} > 0)$	$\mathbb{1}(Earnings_{jt} > 0)$	$\mathbb{1}(Earnings_{jt} > 0)$
Sample	*High-M*	*High-NM*	*Low-M*	*Low-NM*
Baseline Impact	-2.2%	-2.0%	-1.8%	-3.1%
		Gender		
Female	0.9%*	1.0***	0.5*	0.8**
		Race		
Black	-0.1%	-0.4%	0.6%	-0.8%
Asian	-4.1%**	-2.5%**	0.1%	-1.6%*
Other	-1.1%	-0.7%	-1.3%	-2.3%*
		Age		
30-40	2.2%	-0.2%	-1.0%***	-0.4%*
40-50	-6.3%	-0.1%	-1.1%**	-0.3%
		Education		
High	1.2%	1.0%	0.2%	0.6%
¡College	0.9%	0.8%	0.1%	0.7%
¿=College	-0.5%	1.2%*	-0.5%	0.5%
		Income		
Quartile 2	0.1%	-0.4%	0.7%**	0.7%***
Quartile 3	-1.0%	-0.9%*	0.7%	1.2%***
Quartile 4	-2.4%**	-0.9%	-0.5%	0.9%
		Firm Diversification		
Firm	-0.5%	-0.1	-0.5	0.1
		Firm Trading Status		
Export and Import	-0.4%	0.7%	-0.7%	0.4%
Export Only	0.1%	-0.9%	0.5%	0.2%
Import Only	3.9%*	3.0%**	2.2%**	-0.3%
		Firm Size (Employment)		
50-500	0.8%	0.5%	0.2%	0.0%
501-2000	0.8%*	0.4%	0.0%	-0.1%
2001-5000	1.6%*	0.3%	-0.8%**	1.0%
¿5000	-0.6%	0.4%	-0.7%	0.4%

Source: LEHD, LBD and authors' calculations. Each column of each panel of the table displays the implied impact on employment of an interquartile increase in county exposure implied by a triple interaction of *Post* x $County\ Gap_c$ with noted row dummy variables capturing worker attributes. Coefficients are reported in Tables A.6 and A.7. First two rows note the sample and the baseline impact of an interquartile increase in county exposure from the final row of Table 4. ***, **, and represent statistical significance at the 1, 5 and 10 percent levels of the underlying triple DID coefficients.